Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 — Unit 2 · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_02 · Unit 2

1 UE26MA141A · Unit_02: Unit 2

This unit master organizes source material for Engineering Mathematics - I, Unit_02, officially titled Unit 2. It includes 16 video transcripts and 11 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26MA141A · Unit_02: Unit 2

Official catalogue entry: UE26MA141A, Unit_02 (Unit 2). Bucket folder present: yes. Videos: 16; slide decks: 11.

EV0000001

2 Introduction to higher order differential equations, complementary function and particular integral

Study focus: Introduction to higher order differential equations, complementary function and particular integral. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Gita TR.

Key details

- I will be discussing with you higher order differential equations.
- In this unit, we are discussing about higher order linear differential equation.
- An introduction to higher order linear differential equation.

- Next, we will be learning about homogeneous and non-homogeneous linear differential equation with constant.
- We've been learning methods to solve homogeneous and non-homogeneous LDE linear differential equation with constant coefficients.
- We are also discussing about linear differential equation of higher order with variance.
- In fact, we are discussing two different forms of linear differential equation with variable coefficients and methods to solve the two different forms of linear differential equation with variable coefficients.
- We are also learning a method called method of variation of parameters, which is used to solve only second order linear differential equation.
- And we are learning at the end application of higher order linear differential equation to solve problems on electric circuits and simple harmonic motion.
- First is the definition of linear differential equation.
- We are also learning the general form of higher order linear differential equation.
- We are learning different forms of higher order linear differential equation.
- Linear differential equation with constant and with variable coefficients.
- We are learning a very important concept called complementary function and particular integration.
- Why do we have to study differential equations?
- We know that differential equation play a very prominent role in the field of engineering, science, and economics and other disciplines.
- They serve as mathematical models to solve any real world problem.
- They help in modeling every physical, biological and technical process.
- Whether the order of the differential equation is 1 or higher than 1, more than 1, ordinary differential equation is classified into two types based on linearity.
- Non-linear differential equation.
- Is a linear differential equation.
- A differential equation which is not linear is a nonlinear differential equation.
- that this both the equations are of order 1, this is LDE of order 1, and this is nonlinear differential equation of order 1.
- And in the previous sessions, you learned methods to solve both LDE of order 1 and non-linear differential equation of order 1.
- But in the coming session,.
- Sessions we will be talking about only linear differential equation.
- What is x here? x is the independent variable and y the dependent variable which is dependent on x.
- Do you call this differential equation a linear differential equation?

Example or application from the lecture

- Let us first look at the content of this unit.
- Let us just write an example for linear and nonlinear differential equation.
- An example for linear differential equation is dyed by dx plus 2x into y is equal to x square and an example for nonlinear differential equation is y square d by dx plus sine y is equal to x square e to the power 2y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

and its derivative occur in first degree.

Key details

- And are not multiplied together is a linear differential equation.
- Look at this differential equation.
- It is y to the power 1, dy by dx to the power 1.
- Also, the equation do not contain the dependent variable and its derivative in product form.
- Do not have y into dy by dx.
- What is the odd of the differential equation.
- Order of a differential equation is the order of the highest order derivative present in the differential equation.
- It is dy by dx.
- But in the sessions we will be talking about only higher order linear differential equations.
- Definition goes this way.
- A differential equation is said to be linear if the coefficient y and its derivatives are functions of x are constants.
- The degree of y and its derivatives is 1.
- Now, first order linear differential equation.
- We know it is of the form dy by dx plus p into y is equal to q, where p and q are functions of x alone.
- See how does it look like a different equation the form P naught into D2y by Dx2 plus P1 into Dy by Dx plus P2 into Y is equal to capital X where.
- P0 P1 P2 and capital X are functions of the independent variable X and P0 equal to 0 is a linear differential equation of Albert 2.
- Why it is a linear different differential equation?
- What is the definition of linear differential equation?
- A differential equation in which two conditions.
- What is the first condition?
- The coefficient of y and.
- Its derivatives should be functions of x.
- The coefficient of y is p2, dy by dx is p1, and p naught, they are functions of x.
- And also the degree of y and all its derivatives is 1 y to the power 1 d y by dx to the power.
- 1 d2 y by d is 2 to the pi 1.
- Because the order of this differential equation is 2, so it is a second order linear differential equation.
- P0 P1 P2 P n or functions of X alone is a linear differential equation of R n provided P naught not equal to 0.
- The coefficients P naught P1 P2 Pn are functions of x.
- And also the co and also the degree of y.
- It's it all its derivatives, it is dn y by dx n to the power 1.

- n minus one third or derivative of y with respect to x to the power one dn minus two y by dx n minus two to the power one.

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EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

Plus E a n into Y is equal to capital X where E naught even into A N are either functions of X or constants.

Key details

- capital X a function of X.
- What are the two different forms?
- One is linear differential equation with constant coefficients.
- Look at the coefficients of the derivatives.
- It is the coefficient of y is 1, dy by dx is 7, d2 y by dx2 is minus p, d3 y by dx3 is 2.
- What is the order of.
- The differential equation.
- Then the differential equation is linear differential equation with variable coefficient a01 a2 a n are functions of x.
- 1 can be written as 1 into x to the power 0.
- Methods to solve linear differential equation with constant coefficients.
- LDE with constant coefficients can be classified into two types based on this cap right hand side function capital X.
- One is called homogeneous linear differential equation.
- The linear differential equation with constant coefficients in which.
- This capital X right hand side function is zero.
- And the other one is non-homogeneous differential equation.
- If RHS function is a non-zero function of the independent variable, then the differential equation is non-homogeneous LD.
- RHS capital X is 0, RHS capital X is a 10 function of X.
- Equal to 0.
- Because RHS is 0, it is a homogeneous LDE.
- See, I have taken the same.
- Every non-homogeneous linear differential.
- Equation has a homogeneous linear differential equation associated with it.

Example or application from the lecture

- So an example for it, 2d3 by dx3 minus 5, d2y by dx2 plus 7 dy by dx plus y is equal to x square.
- An example for this is 2x cube into d3y by dx3 minus 5x square plus 2 into d2 y by dx2 plus x into dy by dx plus y is equal to x square into sine x.
- So it is an example for linear differential equation of order 3 and with variable coefficients.

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EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

Let us learn about the differential operator.

Key details

- A differential equation.
- What is this differential equation?
- A1, a2, a3, a and say they are constants.
- And linear differential equation with constant coefficient, and it is a non-homogeneous linear differential equation.
- plus a1 d to the power n minus 1 of y plus a2 d to the power n minus 2 of y plus a n into y is equal to capital X.
- You can factorize this f into linear factors, and the linear factors can be written in any order.
- This can be treated like an algebraic expression and it obeys the laws of algebra.
- The linear differential operator.
- The first component is called complementary function, and the second component is called particular invariant.
- And this complementary function is the complete solution of the associated homogeneous linear differential equation.
- That the solution of the whole non-homogeneous linear differential equation has two components.
- One is called complementary function, and the other one is called particular integral.
- And asked to solve the non-homogeneous linear differential equation, you need to find the complementary function and the particular integral of the differential equation.
- and dependence is defined for n number of functions.
- A set of functions f1, f2, f3, fn are said to be linearly dependent on an interval i.
- If there exists constant c1 and c2, c three, cn, not all zeros such that c1 into f1 plus c2 into f2 plus c n into f n is equal to 0.
- What do you mean by this?
- See the linear combination of the functions f1 f2 f n which is c1 f n plus c2 f2 plus c n f n is equal to 0 even for non-zero constant c1 c2 c three c n then the function cf1 f2 f n are said to be linearly dependent functions.
- What are linearly independent?
- If not linearly dependent, the functions are called linearly independent functions.
- The set of linearly independent.
- If the set is linearly independent, so what do you mean by it?

Example or application from the lecture

- We know that the first order differential operator is given by d by ts.
- capital D is the notation given for the first order differential operator D by Dx.
- Now let us first discuss in detail about this complementary function.

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EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

equal to 0 only when c1 c2 c 3 cn 0.

Key details

- That is linearly independent function.
- The linear combination of the function c 1 f1 plus c2 f2 plus c n f n is equal to 0.
- It is true only when the arbitrary constant c1 c2 c3 c n are zeros on that is a trivial case otherwise this is not equal to zero for non-zero arbitrary constants c one c2 c three c n this is not equal to zero if f1 f2 f3 fn are leaving.
- Linearly independent of each other.
- See two functions y now y1 of x and y2 of x are said to be linearly independent when the linear combination c1 into y1 plus c2 into y2 is not equal to 0.
- 0 for non-zero constants c1 and c2.
- That is C C1 by 1 is not equal to minus C2 into Y2.
- That means that y1 not equal to minus C2 by C1 into.
- R y1 is it not equal to minus c2 by c1 into y2.
- Two functions to be linearly independent of each other.
- Neither of the function is a constant multiple of the other.
- That is, y1 cannot be written as some constant into y2, or y2 cannot be written as some constant into y1.
- Then the functions y1 and y2 are said to.
- Be linearly independent functions.
- Like see, this is homogeneous because RHS is 0, K1, K2 are constant, so it is a linear differential equation with constant coefficient.
- How does the general solution of this differential equation?
- See, now it's very important if y1 and y2 are two linear.
- linearly independent solutions of this.
- Second order homogeneous linear differential equation with constant coefficients, then the linear combination C1 by 1 plus C2Y2 is also a solution of the differential equation.
- See, please understand if y1 and y is 2 are linearly independent, then the linear combination C1 by 1 plus C2Y2 is also a solution.
- Why C1 by 1 plus C2Y2 is a solution only when y1 and y2 are linearly independent.
- independent functions, but if y 1 and y2 are linearly independent, then the linear combination c1 by 1 plus c2 by 2 is not equal to 0 for non-zero constant c1 and c2.
- And this is also a solution of the differential equation.
- In fact, this represents the complete solution of the differential equation.
- Very important concept.
- The complete solution of homogeneous linear.
- differential equation with constant coefficient is of the type c1 y 1 plus c2 y2 where.
- Y1 and Y2 are two linearly independent functions.

- n third d n y by dxn plus a1 into n minus one third derivative of y with respect to a x plus a n of y is equal to 0 where a1 a2 a3 a n are constant.

Example or application from the lecture

- Linear differential equation with constant coefficients by first taking an example of now.
- Let us consider a second order LDE with constant coefficients in the type d2 y by dx2 plus k1 dy by dx plus k2 div is equal to 0.

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EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

Core explanation · 25:00–29:59

has n linearly independent solutions y1 by two y three y ninear combinations c1 by one plus c2 by two plus cn by n also represent a solution of the same differential equation and it is the complete solution of the differential equation.

Key details

- And the complete solution of the differential equation is called complementary function of the differential equation.
- The complete solution of linear differential equation, which is homogeneous, and it is of order n will be of the form.
- is equal to c1 by 1 plus c2 by 2 plus cn by n and the complete solution is called complementary function of the differential equation.
- With constant coefficients is called complementary function of the differential equations.
- F of d of y is equal to capital X, where f of d is the linear differential operator.
- One is the complementary solution, other one is a particular solution.
- What do we mean by this first statement?
- Homogeneous linear differential equation.
- The associated homogeneous differential equation is f of d of y is equal to 0.
- Y is equal to YP is a particular solution.
- One solution of the non-homogeneous differential equation f of d of y is equal to capital S, then the complete solution of this differential equation is of the form y is equal to yc.
- C plus YP.
- Why it is called as the particular solution about which we will be learning later in a different session.
- What is Yc?
- It is the complete solution of the corresponding homogeneous differential.
- This particular solution YP is also called particular integral.
- Now in the next session, we will be learning.
- Methods to find the complementary function of the differential equation.

Example or application from the lecture

- C f of d of y is equal to capital X is the given non-homo.

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EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

3 Rules for finding the Complementary Function, Problems

Study focus: Rules for finding the Complementary Function, Problems. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000026 · EV0000027

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

From now on, we will be discussing about only higher order linear differential equations.

Key details

- We will be solving a higher.
- Order homogeneous linear differential equation with constant coefficients.
- We know that the general solution of homogeneous.
- LD with constant coefficients is called complementary function.
- Every homogeneous linear differential equation with constant coefficients has an algebraic equation associated with it.
- The algebraic equation is called auxiliary equation, and the general solution of the.
- Homogeneous linear differential equation depends on the nature of the roots of the auxiliary equation.
- The auxiliary equation has real and different rules, real and distinct rules.
- When the rules of the auxiliary equation are real and equal.
- And when the rules of the auxiliary equation are imaginary.
- Differentiating this y one time and two times with respect to x, we get y dash is equal to em into e power mx and y double dash is m square into e to the power mx.
- You can substitute, you can plug in these equations into the differential equation.
- Only when y is equal to e to the power mx is a solution of the differential equation.
- We know that a to the power mx is not equal to 0 for any real value of x.
- So, which means that m square plus k1m plus k2 equal to 0.
- K1 into capital D plus K2 of Y is equal to 0.
- Or how are we writing the auxiliary equation of the differential equation?
- This equation is f of b.
- of y is equal to 0.
- In f of d you just replace b by m.
- And equate the algebraic expression to 0, we get the auxiliary equation.
- Not a homogeneous differential equation of this type.

- The auxiliary polynomial of this differential equation is f of m.
- What is f of m? f of d is d squared plus k1 into capital D plus K2.
- In F of D replace D by M.
- Linear differential operator with constant coefficient f of d has.
- An auxiliary polynomial associated with it, which is F of M.
- Because the general solution of the homogeneous linear differential equation depends on the nature of the loads of the auxiliary equation.

Example or application from the lecture

- First let us learn how in detail about auxiliary equation.
- For that let us consider a simple second order homogeneous linear differential equation of the type d2 by b dx2 plus k1 d by dx plus k2 of y is equal to 0.

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EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 05:00–09:59

If we are discussing three different cases.

Key details

- So what is the first case when the rules of the auxiliary equation m1 and m2 are real and distinct.
- So because the auxiliary equation is.
- A quadratic equation, it has two roots, m1 and m2.
- Y is equal to some constant c1 into e to the power m1x plus c2 into e to the power m2x.
- The two real and distinct roots of the auxiliary equation.
- What is x?
- X is the independent.
- variable C1 C2 are the arbitrary constant.
- This is how the general solution of the differential equation look like when the roots are real and distinct.
- See what is the order of the differential equation to.
- We have learned in the previous sessions that the general solution of a first order linear diffusion differential equation has one arbitrar constant in its general solution.
- Two arbitrary constants and an n th order differential equation has n arbitrary constants in its general solution because here the order is 2, there are two arbitrary constants in the solution.
- are real and distinct.
- So now first you need to write the auxiliary equation of this differential equation.
- In f of d replace d by m and equate it to 0.
- Another simple way of learning to write the auxiliary equation.
- In this differential equation, you replace y by 1, dy by dx by u.
- D2 y by dx2 by m square.
- If the differential equation has d3 y by dx3, replace it by m square.

- Solve this auxiliary equation, find out the linear factors and find the roots.
- The roots are minus 3 and 1.
- Analyze the roots.
- The roots are real and they are different.
- It is c1 e to the power m1x plus c to e to the power m2x.
- 2x plus c2 e to the power 1 in 2x.
- When the roots of the auxiliary equation are real and equal.
- M1 is equal to M2 is equal to M C.
- Then the general solution, the differential equation is y is equal to c1 plus c2 into x, the independent variable x into e power mx.
- The general solution of the homogeneous linear differential equation of order 2 if the roots of the auxiliary equation are real and equal.
- Y is equal to C1 plus C2x into E2.
- Yes, LDE with constant coefficient.
- And the auxiliary equation is m square plus 4m plus 4 is equal to 0, which is m plus 2 the whole square.

Example or application from the lecture

- If the roots of the auxiliary equation are real and distinct, then the general solution of this second order homogeneous linear differential equation is given by.
- Which is associated with the given homogeneous linear differential equation with constant coefficients.
- So let us illustrate this concept by taking an example.

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EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 10:00–14:59

And minus 2.

Key details

- The two rules are real and repeated.
- It is c1 plus c2 into x into e to the power em x.
- What is m over here?
- X is the solution of this homogeneous differential equation.
- Third case is when the roots are complex imaginary.
- Let the roots be m1 is equal to alpha plus i beta, m2 is alpha minus i beta.
- We know that the complex roots always occur in pairs, conjugate roots occur in pairs.
- complex is y is equal to e to the power alpha is c1 cos beta x plus c2 sine beta is understand this it is e to the power alpha is the real part of the root and beta is the imaginary part of the root so it is e to the alpha into independent variable x.
- into C1 arbitrar constant C1 cos beta x plus c2 sine beta s.
- equation are complex in nature.
- That is the auxiliary equation.

- Using the quadratic formula, find the roots of this equation.
- It is minus b plus r minus square root of b square minus 4ac by 2 into a.
- A is 1, B is minus 4, and C is 13.
- Substitute in the quadratic formula.
- Get the roots.
- The roots are 2 plus 3i, 2 minus 3i.
- Alpha is 2 and beta is 2.
- Because the roots of the auxiliary equation are complex.
- C1 cos 3x plus c2 sin 3x.
- This is the general solution when the roots are complex in nature.
- To an n thirder homogeneous linear differential equation.
- So this is the auxiliary equation because the degree of this algebraic equation is n.
- How many roots?
- The general solution is just extend the same concept which we learned for a second order homogeneous differential equation to an nth order differential equation.
- Plus C and E to the power mx.
- What is it when the roots m1, m2?
- M3 M and all are real and equal.
- Say equal to M1 equal to M2 equal to M3 equal to M and equal to M.
- be C1 plus C2X plus C3 X square plus C n x to the power n minus 1 into e to the power m the equal root mx.

Example or application from the lecture

- Now let us consider an example d2y by dx2 minus 4 into dy by dx plus 13y is equal to 0.
- Let us consider different possibilities.
- Let us consider.

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EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 15:00–19:59

You are stopping at x to the power n minus 1.

Key details

- If the order of the differential equation is 3, say you are stopping at c3 x to the power 3 minus 1, which is x to the power 2.
- If the order of the differential equation is 2, you are stopping at C2 x to the power 2 minus 1, which is x.
- Look at the number of arbitrary constants in the general solution.
- How many arbitrary constant in an nth order differential equation?
- See, there are n constant.
- C1, C2, C3, C.
- And the remaining n minus 4 roots are real and distinct.
- So the first four roots are real and equal to same m.

- Remaining n minus 4 are real and distinct.
- What is it?
- If there are two pairs of imaginary roots, alpha plus r minus i beta, gamma plus r minus i delta, and the remaining n minus 4 roots are real and distinct.
- See there are totally n roots.
- The first two roots are alpha plus i beta and alpha minus pi beta.
- The next two roots are gamma plus i delta, gamma minus i delta.
- The remaining n minus 4 are real and distinct.
- What is it?
- E to the power alpha x c1 cos beta s plus c2 sin beta x.
- Complex roots e to the power gamma x into c3 cos delta x plus c4 sine delta x and for the remaining n minus 4 roots it is c phi e to the power m phi x.
- Plus C6 e to the power mc plus c n e to the power mnx.
- Here I should write the constants as C3 and C4.
- It is written as C and C2.
- We have to change this.
- What is the other possibility?
- And the remaining n minus 4 roots say they are real and distinct.
- into C1 plus C2X for the repeated complex load C1 plus C2X into cos beta x plus c three plus c4 x into sine beta x.
- into for the remaining n minus four roots it is c phi e to the power m phi x plus c six e to the power m c plus cn e to the power m n x.
- If the complex roots are repeated in the auxiliary equation, then we write this way C1 plus C2x into cos beta x plus c3 plus c4 x into sin beta x.

Example or application from the lecture

- Let us say the first four roots, m1, m2, m1.
- differential equation see for the first four equal roots let us write it is c1 plus c2x plus c3x square plus c 4x to the power 3 because the first four roots are real and equal into e to the power mx.
- Let us discuss one more possibility.

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EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 20:00–24:59

Homogeneous linear differential equation with constant coefficients.

Key details

- A third order homogeneous linear differential equation.
- Dq y by dx q plus 4 d by dx is equal to zero.
- What is it?
- It's a homogeneous third order LT with constant coefficients.
- General solution of this differential equation called complementary function.

- What is the auxiliary equation?
- F of D equal to when equate.
- f of m to zero.
- See this linear differential operator.
- For every linear differential operator, we can introduce a characteristic polynomial mq plus 4m and the.
- And when mq plus 4m is equated to 0, we get the auxiliary equation.
- I have taken a simple cubic equation.
- 3 roots, because the degree of this equation is 3.
- to zero m is equal to zero and m is equal to plus or minus divide.
- Here C 0 for the real root I'm writing C1 E to the power 0x.
- sine to x does the solution of this c one plus c2 cos to x plus c three sin to x.

Example or application from the lecture

- Let us solve.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

4 Particular integral of standard functions - Type 1

Study focus: Particular integral of standard functions - Type 1. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000046 · EV0000047

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Methods to solve a non-homogeneous linear differential equation of the type f of b of y is equal to capital X when capital X is equal to E par ax.

Key details

- In the previous session, we learned methods to solve.
- Homogeneous linear differential equation with constant coefficients.
- We know that the general solution of the non-homogeneous LDE with constant coefficient has two components, complementary function and particular integral.
- And complementary function is the general solution of the associated homogeneous differential equation.
- We already learned methods to find complementary function.
- Basically, in this session, we are finding methods to find.
- The particular integral of the differential equation f of df y is equal to e power ax.
- What is this inverse differential operator?
- Why do we need this inverse differential operator?

- The inverse differential operator is denoted as 1 by eight.
- f of d.
- 1 over f of d and 1 and f of d are inverse operators and f of d of 1 over f of b of x is equal to x itself.
- They are inverse operators.
- You can change the order.
- You can also write it as 1 over f of d of f of b of x is equal to capital X.
- over E f of D or 1 over Af of B of X.
- It's very important statement.
- F of B of Y is equal to capital X.
- It is a non-homogeneous linear differential equation.
- Just recall what is a non-homogeneous LD?
- If RHS of the differential equation is a non-zero function of the independent variable, then the differential equation.
- Is called as a non-homogeneous linear differential equation.
- Is equal to x itself.
- Why it is equal to x?
- Because by definition of 1 over f of d, 1 over f of d of x is that function of x, which when operated upon by f of d gives back x.
- So y is equal to 1 over F of D of capital X satisfies the equation f of d of y is equal to capital X, which means that y is equal to 1 over F of D of X.
- Is a solution of the differential equation.
- And this solution is called particular solution of the differential equation f of d of y is equal to capital X because this 1 over F of D of X do not contain any R.

Example or application from the lecture

- First, let us learn a very important concept called inverse differential operator.
- Let us prove this.
- Let us prove this statement.

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EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 05:00–09:59

If you have a non-homogeneous differential equation here of Df y is equal to capital X, we know that the general solution of this non-homogeneous equation has two components.

Key details

- One is called complementary function, which is abbreviated as CF, and the other.
- It is the complete solution of the corresponding homogeneous differential equation.
- Now the next question is how do we find 1 over F of D of capital X?
- What is the meaning of this?
- It is 1 over F of D of capital X when F of D is equal to simple capital D.
- with respect to x without writing the constant of integration.

- 2 times integration of x without writing the constant of integration.
- 1 over D minus Ef capital X, so it is 1 over F of D of capital X, where F of D is the linear factor d minus E.
- What is 1 over d minus a f capital X?
- Replace a by minus a in the formula.
- Integral, E power X, X, Dx.
- It can be simplified in two ways.
- One way is applying the operator one by one.
- First you apply the operator 1 over d minus b on x.
- And you then you apply the operator 1 over d minus e on the resultant function of x.
- You get a function fx.
- Then on the resultant function you apply the operator 1 over d minus a.
- This is one way of simplifying 1 over d minus a into d minus b of x.
- The other way is resolve this 1 over d minus e into d minus p into partial fractions.
- Then use this formula and simplify.
- This 1 over f of d is an operator which is acting on the function x.
- It is not x into 1 over f of b because this is an operator which is acting on the function x.
- What if f of d is a cubic equation, cubic expression, then f of d would be product of three linear factors.
- Then this method is very then the now we are discussing methods to find particular integral when capital X, which is the right hand.

Example or application from the lecture

- Component is particular integral which is given the notation PI.
- We learn methods to find complementary function and the particular integral is given by 1 over f of d of capital X.
- So let us first learn some results, important results wherein f of d takes a simple function of D.

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EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 10:00–14:59

Side function of the differential equation is a specific function of x.

Key details

- The first case being.
- When capital X is equal to E par ax.
- Basically, we are now learning methods to find the particular integral of e par ax, an exponential function a par ax.
- is e par ax by f of a provided f of a not equal to 0.
- It is 1 over f of b of e po is e power a x by f of a.
- In f of d, just replace d by a to get f of a.
- Provided f of a not equal to 0, the denominator not equal to 0.

- It is e power minus ax by f of minus a in the linear differential operator re.
- Place d by minus a if you have e par minus a x as the function.
- The cases of failure.
- When is f of a equal to 0?
- When d minus a is a factor of f of d.
- Then f of a is equal to 0, then a becomes a root of the equation f of d and f of a is equal to 0.
- When f of a is equal to 0, the formula is 1 over f of d of e power e x is equal to x into.
- E par a x by f dash a provided f dash a not equal to 0.
- See x has to be written outside the operator.
- It is x into 1 over f dash d of e par ax.
- And in f dash d replace d by a.
- You have f of d differentiate with respect to d to get f dash d.
- Then in f dash d put d is equal to a to get f dash a.
- C please understand x has to be written outside the operator x to the power 1 into e power a x by f dash a.
- What if f dash a equal to 0?
- When d minus a and d minus a two linear factors of f of b.
- into e par a is by f double dash a provided the denominator f dash double dash e not equal to zero.
- You differentiate this f of d two times with respect to d and then substitute d is equal to a in the second derivative to get f double dash a so provided this is not equal to 0.
- What if this f double dash a is equal to 0?
- The formula is x cube into e power e x by f triple dash a and so.
- But the same formula is applicable for the following functions.
- So, what is the first one? x is equal to k constant.
- D by 0 it is k by f of 0.
- 0 is not a root of the equation f of d is equal to 0.
- If 0 is a root, then it is x in 2.
- K by f dash 0.
- If f dash 0 is also 0, x square into k by f double dash 0 and so on.
- Now, so the same formula is applicable to next sine hyperbolic eggs.
- What is sine hyperbolic A in terms of exponential function?
- It is sine hyperbolic A is E par Ax minus e power minus e x by 2.

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EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 15:00–19:59

minus e power minus a x by f of d.

Key details

- Minus e par minus e x by f of b.
- In the first term we place d by a and in the second term replaced d by minus a.

- A now another function cos hyperbolic x.
- What is cos hyperbolic x? e par e x plus e par minus e x by 2.
- And what is cos hyperbolic x by f of d?
- It is half into e par a x by f of a plus.
- A power minus a is by f of minus a.
- There is one more function. x is equal to a power x.
- How is a power x written in terms of exponential function?
- can be written as e par log a par x.
- It is e par log.
- E per log a par x, that is e per log a into x.
- It is also a constant which is log a.
- Replace d by log a.
- V plus d by minus login if you have a bar minus.
- It is a non-homogeneous LDE.
- Why it is a non-homogeneous LDE?
- Because RHS is not equal to 0.
- What is the order of the differential equation?
- And this differential equation is a differential equation with constant coefficient.
- Solving this involves two major steps.
- The first step is to find the complementary function, and the second step is to find the particular integral.
- First, to find complementary function, I need to write the associated homogeneous differential equation of this differential equation.
- It is d square minus 2d plus 2 of y is equal to 0.
- And the auxiliary equation of the homogeneous equation is m square minus 2m plus 2 equal to 0.
- The roots are 1 plus r minus i because this is a quadratic equation, two roots, one root is 1 plus i and other root is 1 minus i.
- When the roots of the auxiliary equation are complex in nature.
- C1 cos what is beta here?
- The coefficient of i is 1.
- C1 cos 1x plus c2 sin x.
- Non-homogeneous differential equation is this.
- To find the particular integral.
- Particular integral is e power minus x.
- by d square minus 2t plus 2 plus e power x by d square minus 2t plus 2 because the RHS function is a power minus x plus e po x, which I split it into two terms.
- It's simple.

Example or application from the lecture

- Let us solve a problem.
- Y c the complementary function is also given the notation y c it is subscript c y c is equal to e power alpha x that is e to the power 1 into x into.
- The complementary function of the given.

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EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 20:00–24:59

replace d by e to the power a is one.

Key details

- Replace d by one.
- How do you simplify the first term?
- Replace d by minus log a.
- Minus 2 into minus log a plus 2.
- Plus e power x in this replace d by 1.
- equation is Y C plus 1.
- This is how we find out the particular integral of the non-homogeneous differential equation when capital X the RHS of the differential equation is an exponential function.

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EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

5 Particular integral_Type 1 and Type 2

Study focus: Particular integral_Type 1 and Type 2. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000066 · EV0000067

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we solve problem on finding the particular integral when capital X is k power X and X equals.

Key details

- Sine of AX plus B or cos of AX plus B.
- You have already learned how to find Pi when right hand side capital X is e power AX, where a can be positive, negative, or even zero.
- over f of a it means we will replace d by a in f of d provided f of a has to be not zero in case f of a becomes zero as from theory the particular integral is x.
- Expressed in this particular form that is x times e power ax divided by f dash of a.
- It means we differentiate f of d with respect to d once and again replace d by a again f dash of a should be not zero.
- In case f dash f a becomes zero, so the particular integrity will be modified in this particular form.

- That is x square times e power a x divided by f double dash a and so on.
- Observe carefully the right hand side is one.
- This is nothing but e power 0x where a is taken as 0.
- Replace d by 0, we get 0 plus 0 plus 5, so 1 by 5.
- So because this is non-zero, we can stop here.
- P particular integral PI write the formula one over.
- D square plus four d plus four times a power minus two x.
- Observe carefully, we need to replace d by a when a is minus two.
- Again we see that denominator is zero when d is minus two.
- Therefore this is written as x square times one over two d becomes two four becomes zero e power minus two x.
- The right hand side is sine hyperbolic X.

Example or application from the lecture

- Let us solve problems.
- Take an example of finding the particular integral for this different equation.
- Let us take another example where the right hand side is a power minus 2x.

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EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 05:00–09:59

Therefore this is same as e power x minus e power minus x.

Key details

- As you can see this is P1 and P2.
- Even in the second stage, f dash f is also zero.
- Therefore, P becomes x square times e power x divided by 6.
- Whereas in the.
- Second case denominator is non-zero.
- Finally Pi is P1 minus P2.
- When x is sine of a x plus b or cos of a plus b.
- Remember, if we want to apply this case, there are certain constraints.
- F of D should have D square in it.
- Constraint number two.
- We replace D square by minus A square.
- Remember this is not minus A whole square, it's minus A square.
- to get d square in f of d.
- That is why when you write the particular integral this is not f of t this is f of d square.
- And we are replacing d square by minus a square, provided this has to be non-zero.
- In case it becomes zero, we go by case one.
- Introduce one x differentiate f of d bet.
- Respect to D once with respect to D1s.

- Again, that derivative should have d square, then replace d square by minus a square, and so on.
- Again this is minus a square.
- Find the particular integral for this differential equation where the right hand side is cos of row 3x.
- And you can see a is root three.
- Minus three whole square.
- Therefore this is minus three.
- Therefore denominator becomes zero.
- In this case Pi is expressed in the form x into derivative of d squared becomes two d.
- Take x by two side integrate cos root 3x with respect to x, then we arrive at the result.
- X by 2 sine root 3 x divided by root 3.
- So this problem should be solved by both case 1 and case 2 because for this one we use e ball 0x and for this we use case number 2.

Example or application from the lecture

- Let us apply case one two times.
- Let us move on to second case.
- Let's say let us take an example.

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EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 10:00–14:59

It's very easy to see that when you compute d1, when you replace d by gate, denominator becomes zero.

Key details

- Then we introduce one x.
- I differentiate with respect to D this 3D square.
- This is minus 40 and this is 1.
- Again 3d square becomes 0 and 4d becomes 0, the left out number is 1.
- Therefore, the answer for this P1 is simply x by 2.
- How do we solve P2?
- And you observe carefully as per theory part this f of d doesn't have d square.
- It means multiplying divided by the conjugate of 8 minus 3d.
- It means 8 plus 3d of cost 2x divided by 8 minus 3d into 8 plus 3d.
- That's going to be 64.
- Minus 36 B square.
- In the denominator,.
- When you replace d square by minus 4, you get 100.
- Sine of Ax plus B or cos of X plus B.

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EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

6 Particular integral of standard functions - Type 2

Study focus: Particular integral of standard functions - Type 2. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000078 · EV0000079

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In today's sessions, we are learning a method to solve a non-homogeneous linear differential equation of the type f of df is equal to capital X when capital X is trigonometric function of the type sine a x plus b or cos a x plus b.

Key details

- If the resultant function still contains any d term, then convert those d terms into d square terms and then replace d square by minus into a square.
- And d write as it is.
- D square replace it by minus into a square.
- And d cube write it as d square into d.
- Replace only d square by minus into a square.
- For cos a x plus b is cos e x plus b by f of d square is cos e x plus b by f of minus into a square provided the denominator f of minus a square not equal to 0.
- Now the next question is what if the denominator f of minus a into a square equal to 0.
- When b square plus a square is a factor of f of b.
- Then f of minus a square will be equal to 0.
- Sine a x plus b by f of d square is x into sine a x plus b by f dash of minus a square provided the denominator f dash of minus a square not equal to 0.
- C x has to be written outside the operator.
- x into sine a x plus b by f dash of minus a square.
- And in f dash d, you replace only d square terms by minus into a square.
- If that function still contains any d term, then convert those d terms into d square terms and then replace d square by minus a square, provided the demo denominator not equal to 0.
- It is cos a x plus b by f of d square is x into cos a x plus b by f of minus into a square where the denominator not equal to 0.
- Now the next question is what if f of minus f dash of minus a square equal to 0?
- The rule continues if f dash of minus a square equal to zero, then sin a x plus b by f of d square is x square.
- into sine a x plus b by f double dash of minus a square where in f double dash of minus a square not equal to 0.

Example or application from the lecture

- I'll just give you an example.
- Suppose say f of d, the given function of d is say d plus d square plus dq.
- In the given f of d, differentiate it with respect to d to get f dash d.

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EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 05:00–09:59

rule continues if f double if double dash of minus a square is 0 then it is x cube into sin a x plus b by f triple dash of minus a square and so on.

Key details

- But we can use the same rule for the following functions also.
- Here you have product of two trigonometric functions.
- Convert the product of trigonometric functions into sum or difference by using transformation formulae.
- See, we know that sine A into cos B.
- is 1 by 2 sin a plus b plus sin a minus b.
- Using the transformation formulae, convert the product of two functions into summary of the trigonometry functions and use the root.
- Using the trigonometric identity.
- What is sin square ax?
- It is 1 minus cos 2x by 2.
- Cos square ax 1 plus cos 2x by 2.
- Using the trigonometric identity, you just write sin square or cos square in terms of cos e.
- Ax or sine Ax and then use the same rule.
- This so same rule applicable for sine cube Ax or cos cube Ax using the same using the trigonometric identity.
- It is a non-homogeneous linear differential equation.
- Because of cos 2x plus 3.
- Non-zero function on the right hand side of the differential equation.
- It is a linear differential equation.
- What is the order of the differential equation?
- We know that.
- The general solution of this non-homogeneous differential equation has two components, the complementary function and the particular integral.
- What is the first step?
- Step one is to find the complementary function.
- So the auxiliary equation is mq plus 1 is equal to the equal to 0 because f of d here is dq plus 1 replace d by m and equate it to 0. mq plus 1 is equal to 0.
- M plus 1 into M square minus M plus 1.
- What is the formula used here?
- A Q plus B cube.

- What is AQ plus B cube?
- A plus B into A square minus A B plus B square.
- To factorize this to find the roots of this, use the quadratic formula minus b plus or minus square root of b square minus four ac by two a and find the roots.
- The roots are m is equal to minus one and half plus or minus root three by two into i.
- The first root is a real root.
- And the other two roots are complex in nature.
- For the real root.
- Alpha here is half and beta is root 3 by 2.
- It is e to the power alpha x into.
- A C2 cos beta x plus c3 psi beta x.
- 1 by 2 into x into c2 cos root 3 by 2 into x.

Example or application from the lecture

- Now let us understand this method in a simple way by taking an example.
- Consider the differential equation dq plus 1 of y is equal to cos 2x plus 3.
- Let us first analyze this differential equation.

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EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 10:00–14:59

plus C3 sin root 3 by 2 into x.

Key details

- It is 1 over f of d of cos 2x plus 3.
- by f of minus into a square provided f of minus into a square not equal to 0.
- Because a here is 2.
- Cause 2x plus 3 by C D I have written as it is, and D square is replaced by minus into 2 square.
- See please note here it is not minus 2 the whole square.
- It is minus into 2 square, which is minus 4.
- See, I told you if there is any D term remaining, D term has to be converted into D square term.
- How do we convert this 1 minus 4D into D square term?
- Multiply the numerator and denominator by its conjugate.
- What is a conjugate of 1 minus 4d?
- It is 1 plus 4d.
- Multiply the numerator and denominator with 1 plus 4d.
- See you please look here.
- See this 1 plus 4d or 1 over, 1 plus 4d are 1 over, 1 minus 4 d are all operators.
- I'll write first I'll simplify the denominator.
- What is that rule?
- We place d square by minus into 2 square which is minus 4.

- That is minus into minus become plus 64 plus 1 65.
- Cas 2x plus 3.
- What do you mean by d?
- It is the first order differential operator d by dx.
- It is minus 2 into sin 2x plus 3.
- Just look at this problem.
- A non-homogeneous linear differential equation with constant coefficients and the order of the differential equation is 4 because the equation has d square into d square, d to the power 4.
- What is d to the power 4?
- It is fourth order differential operator d4.
- 4 by dx to the power 4.
- Now, because it is a non-homogeneous linear differential equation, we need to find the complementary function and the particular integral.
- The first step is to find the complementary function.

Example or application from the lecture

- The particular integral is given by cos of 2x plus 3 divided by dq plus 1 because f of d is dq plus 1.

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EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

Core explanation · 15:00–19:59

M minus 1 the whole square into m square plus 1 equal to 0.

Key details

- How did I get this?
- You just have to replace d by m.
- And equate the expression to 0.
- That is the auxiliary equation.
- There are 4 roots and the roots are.
- 1 comma 1 comma plus r minus i.
- Why these two 1 and 1 roots?
- Because m minus 1 the whole square is 0.
- M minus 1 into m minus 1 0.
- 1 comma 1 are the two roots here.
- M square is equal to minus 1.
- M is equal to square root of minus 1, which is plus r minus i.
- The first two roots are real and equal, and the next two roots are complex in nature.
- The complementary function is c1 plus c2x into e to the power 1x.
- This is for the first two equal roots.
- Next for this complex roots.
- The second step is to find the particular integral.

- Now I have written particular integral as e power x by f of d plus car square x by 2 by f of d because it is e power x plus car square x by 2 by f of d.
- I call this as particular integral 1 and particular integral 2 and simplify them separately.
- Pi 1 plus PI2 where PI1 is a first term and P i2 is a second term cos square x by 2 by d minus 1 the whole square into d square plus 1.
- To find particular integral and what is it?
- E power x by d minus 1 the whole square into d square plus 1.
- How do I simplify this?
- I'm using rule 1.
- What is that rule 1?
- E power e x by f of d is e power a x by f of a.
- Provided f of a not equal to 0, but it's very clear from this d minus 1 the whole square 1 comma 1 are the two roots.
- If I directly substitute d is equal to 1, I get 0 in the denominator.
- So what I do is first I apply the operator 1 over d square plus 1 on e par x.
- And then I'll apply the operator 1 over d minus 1 the whole square on the resultant function of x.
- Replace b by.
- What is a here?
- That is 1 by 2.
- How do I find this now if I replace d by 1 I get f of a 0.
- If I so if I differentiate it again, again I am getting 0.
- So this is equal to first let me differentiate it one time.
- It is x into e power x by.
- 2 into 2 into d minus 1, which is 4 into d minus 1.
- Why did I write x?
- The rule says if f of e is 0, e po ax by f of d is x into e par ax by f dash d and in f dash d replace d by 1.
- X square into E power x by 4 into D minus 1.
- Differentiation is 1.

Example or application from the lecture

- See in the given differential equation f of d is d minus 1 the whole square into d square plus 1.
- Let us solve this algebraic equation because the degree of this algebraic equation is 4.
- Add the two to get the particular integral of the given differential equation.

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EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 20:00–24:59

It is ca square x by 2 by d minus 1, the whole square by d square plus 1.

Key details

- I don't have any formula for ca square x by 2 by f of d.
- That is I cannot directly find particular integral of ka square x by 2.
- I have formula for cos a x or sine x, but not for cos square x by two.
- I use trignometric identity and write cos square in terms of cos ax or cos bx.
- Cos x by 2.
- What I do now it here in the numerator it is the sum of two terms.
- I split it into two terms.
- It is 1 by 2 into d minus.
- 1 the whole square into d square plus 1 plus half into cos x by d minus 1 the whole square into d square plus 1.
- Which rule I am using?
- I'm using rule 1.
- Because 1 can be written as 1 into e power 0x.
- Replace d by 0.
- What is that rule?
- Cos a x by f of d replace d square by minus into a square.
- I apply the operator 1 over d minus 1 the whole square on cos x first.
- And I'll apply the operator d square plus 1 on the resultant function of x.
- So it is first term is 1 by 4 plus 1 by 2 into d square plus 1 into cos x by d minus 1 the whole square.
- What is it? d square minus 2d plus 1.
- How do I simplify this?
- replace d square by minus into a square.
- D term right as it is.
- So what I do is first I simplify the term inside the bracket and then apply the operator 1 over d square plus 1.
- It is integration of cos x with respect to x.
- without writing the constant of integration.
- It is sine x.
- How do I calculate this?
- I use rule sin a x plus b by f of d replace d square by minus into a square I get minus 1 square plus 1 which is 0.
- When you are writing x here you have to differentiate the denominator.
- Differentiation of d square is 2d.
- So for this I need not need not have to use any rule because 1 over d of sin x, that is integration of sine x with respect to x, which is minus cos x, so it is 1 by 4 plus x into cos x by 8.
- It is particular integral 1 plus particular integral 2 and.
- The general solution y is equal to the complementary function plus particular integral.
- See if they ask you to find only the particular integral, you can write up till here.
- If they ask you to solve the differential equation, you should mention this.
- Y is equal to complementary.
- function plus particular integral.

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EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

7 Particular integral of standard functions - Type 3

Study focus: Particular integral of standard functions - Type 3. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000098 · EV0000099

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We are solving a non-homogeneous differential equation of the type f of d of y is equal to capital X, where capital X is equal to x to the power n and is a positive integer.

Key details

- In the previous sessions, we learned methods or rules to find the particular integral of exponential function e to the power ax or trigonometric functions of the type sine ax plus b.
- R cos c plus b.
- So the step the first step being in f of d.
- Take out the lowest d degree common to get the expression in the form 1 plus some function of d which I call it as phi of d or 1 minus phi of d.
- And you are when you are expanding you are writing the terms up to d to the par n because the d to the power n plus 1 of x to the power n, that is n plus 1 third derivative of x to the po n is anyway 0, n plus 2 third.
- Derivative of x to the power n is 0.
- Or 1 minus 5 of d to the power minus 1, you write the terms up till d to the power n, neglecting the higher powers of d.
- Then operate on the numerator term by term by taking capital D is equal to the n to d by dx, d square as d2 by dx to d three d cube as d cube by dx cube and so on.
- The following expressions will be helpful to expand 1 plus phi of d to the power minus 1 or 1 minus phi of d to the power minus 1 in ascending powers of d.
- What is the first equation?
- 1 plus x whole to the power minus 1 is 1 minus.
- Minus x, plus x square, minus x cube, and so on.
- Here in the series, plus and minus sign alternates.
- We need not have to remember that.
- 1 minus x whole to the power minus 1 is 1.
- Minus x, which is plus x.
- Plus minus x the whole square, which is plus x square.
- Minus into minus x cubed, which is plus x cube, and so on.
- Similarly, the next expression may also be used.
- to expand this 1 plus 5 d to the power minus 1 or 1 minus 5 d to the power minus 1.

- 1 plus x whole to the power minus 2, which is 1 minus 2x plus 3x square minus 4x cube.
- See look here in this series also plus and minus sign alternates in the C.
- Then 1 minus x whole to the power minus 2.
- We need not have to remember this.
- So if we remember the first and the third, it is sufficient.
- So the second and the fourth can be derived from the first and the third.
- Find the general solution of d2y by dx2 minus y is equal to 5x minus 2.
- Look at the right hand side function is an algebraic function, 5x minus 2.
- So now because the equation is a non-homogeneous linear differential equation, so the general solution has two components: complementary function and particular integral.
- So the first step being the to find the complementary function.
- For that, first let me first write this in operated notation form capital D square minus 1 of y is equal to 5x minus 2.

Example or application from the lecture

- First particular integral is given by 1 over f of d of x to the power n.
- Let us consider a question.

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EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 05:00–09:59

M square minus 1 is equal to 0 because the degree of this equation is 2.

Key details

- 2 roots, the two roots are plus or minus 1.
- What is the first step?
- Take out the lowest d degree term to get the expression in the form 1 plus phi d or 1 minus 5 d.
- Because minus 1 can be written as minus 1 into d to the power 0.
- Minus of 1 minus d square to the power 1 minus d square.
- That is the first step.
- I wrote it in the form 1 minus 5 of d to the power minus 1 of the function 5x minus 2.
- Now what is the next step?
- Using binomial theorem, expand this in ascending powers of D.
- 1 minus x whole power minus 1, which is 1 plus x plus x square plus x cube.
- Use that formula and expand it in ascending powers of d.
- I get it as minus of 1 minus d square plus d to the power 4.
- See this is 1 minus x whole to the power minus 1, which is 1 plus x plus x square plus 1.
- Before it was 1 minus x, so that is wrong.
- 1 plus x square plus x to the power 4 plus on because it is 1 minus x whole raised to minus 1.
- When I simplify this, see 1 into 5x minus 2 is 5x minus 2, and d square of 5x minus 2.
- What is d square of 5x minus 2?

- 2 times differentiation of 5x minus 2 with respect to x.
- First time differentiation of this 5x minus 2 itself is a constant.
- First order derivative of 5x minus 2 is 5.
- If I differentiate the constant again, I get 0.
- So this simplifies to minus into 5x minus 2 because these when I operate these d terms.
- terms on 5x minus 2 I am getting 0.
- What is particular integral?
- It is 2 minus 5x.
- Therefore, the general solution is y is equal to the complementary function which is denoted as yc plus the particular integral y p.
- Find the particular integral of d square plus phi d plus 4 of y is equal to x square plus 7x plus 9.
- See in this question I need not have to find the complementary function because the question is only to find the particular integrals.
- Considered here an algebraic function of degree 2.
- What is the first step?
- In f of d you take out the lowest d degree term as common factor and get this expression in the form 1 plus 5d or 1 minus 5d.
- You if you take 4 outside, it becomes 4 into 1 plus d square plus 5 d by 4.
- See this is what we get.
- Of x square plus 7x plus 9.

Example or application from the lecture

- Particular integral is given by x square plus 7x plus 9 divided by d square plus by d plus 4.

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EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 10:00–14:59

If you look at the expression over here, algebraic expression, it is of degree 2.

Key details

- minus this term that is d square plus phi d by four plus.
- X square that is d square plus y d by 4 the whole square.
- I am neglecting the higher power sub d because d3, d4, dn of x square plus 7x plus 9 is anyway equal to 0.
- The whole square.
- First use a plus b the whole square and you simplify.
- 1 minus d square by 4 minus 5 b by 4 plus in this a plus b the whole square.
- A square plus 2ab plus b square.
- When I differentiate, I get d square of x square plus 7x plus 9 as constant.
- Higher powers of d is anyway, when it is operated on x square plus 7x plus 9, I get 0.
- I have written only d square by 4 the.
- Sorry, 5 d by 4 the whole square, which is 25 d square by 60.

- I've neglected d square the whole square, that is d to the power 4 into 2 into 5 d into d, that is 10 d cube that I have neglected and I have considered only 5 d by 4 the whole square which is 25 by 5 25 d square by 16.
- See d of x square plus 7x plus 9 that is 1 time differentiation 2x plus 7.
- 2 times differentiation is 2.
- Higher derivatives will be anyway equal to 0.
- 1 by 4 into 1 into x square plus 7x plus 9 minus 1 by 4 into what is d square of this?
- It is equal to 2.
- Minus 5 by 4 into d of x square plus 7x plus 9 is 2x plus 7.
- Plus 20 by 16 into d square is 2.
- This is how we find the particular integraph of integral of algebraic function of the type x to the power n.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

8 Particular integral -Type 3 and Type 4

Study focus: Particular integral -Type 3 and Type 4. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000110 · EV0000111

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will solve problems on finding the particular integral when x is a polynomial, that is x is x power m and.

Key details

- When x is epoch a x into v, where v is a function of x.
- As usual, in this case the particular integral is expressed in the form one over f of d times x power m.
- On all f of b is written as f of b inverse times x power m.
- Once we do this, we expand this in ascending power sub b using binomial series.
- Up to the term containing d power m it means the terms beyond d power m are neglected because.
- P power n of x power m is zero for n is greater than m.
- It means n is three and m is two.
- And you can clearly see n is better than m and it's going to be zero.
- When n is one, when you differentiate first time, x square become two x, seven x becomes seven.
- When n is two, two x becomes two, seven becomes zero, when n is three,.
- It becomes zero.

- It means the nth derivative of x power m is zero when n is m.
- So because of this condition, it comes beyond b power m or neglected.
- Solve the differential equation d square plus 3d plus 2 of y equals x power 3 plus x square.
- You can observe carefully the value of M is three here.
- The axillary equation is m square plus 3m plus 2 0.
- The roots are minus 1 and minus 2.
- plus c 2 times e power minus 2x.
- It is 1 over the d square plus 3d plus 2 of x square plus x square.
- from theory part, when we apply case number three, it is convenient to express f of d either in the form of 1 plus d or 1 minus d, so that we can apply binomial series.
- It is written as 2 of 1 plus d square plus 3d by 2.
- It becomes 1 plus 3d, 1 plus d square plus 3d by 2, something like minus 1.
- from logic whenever we have one plus it is alternative sign on the right hand side one minus x plus x square minus x power three and so on.
- And remember x is nothing but d square plus three d by two.

Example or application from the lecture

- Let us start with case number three when x is x power m.
- Let us take an example.
- Let us take x square plus seven x plus eight.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 05:00–09:59

This is x.

Key details

- This is x square.
- This is x power three and so on.
- Expanding this using binomial series up to the term containing d power m.
- Logically speaking, from x power 3 plus x square.
- The highest power of x is 3.
- It means when we use binomial series we should go only up to d power 3.
- The term beyond d power 3 are neglected, it means d power 4, d power 5, so on they're all neglected because anyway they become zero.
- Obviously this is considered, D square is considered, 3D is considered because they are less than 3.
- When you look at d square plus 3d whole square.
- We don't want a square.
- We want b square.
- We want 9 d square.
- And we don't want yeah we want 2ab that is 6d q.
- What about the fourth term?

- We don't want a power 3 but we want bq because it's 27 d power 3.
- If you have d of that, differentiate x power three plus x square once with respect to x.
- If it is d square, differentiate two time.
- If it is d power three, differentiate three times.
- The final PI is going to be half times x power 3 minus 7x square by 2 plus 30x by 4 minus 31 by 4.
- very well how to compute the complementary function.
- The roots are minus 1 plus r minus root phi by 2 into x.
- Then what about particular integral?
- Minus 25 by 9.
- Where b is a function of x.
- Remember, this is a combination of case four plus case three and case two.

Example or application from the lecture

- Let us look at another example. x square plus 2x minus 1.
- Again the task is to reduce this polynomial to one plus x mar let us take three common you get one plus two d square plus two d by three and this part looks like even x.
- Let us move on to case four.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

Core explanation · 10:00–14:59

Depends on B.

Key details

- This is expressed in the form e po x times we choose a from here, replace a d by d plus c in f of b that is here.
- then retain v of x as it is.
- From here to here depends on case four it could be case two and it could be case three.
- If v is sign or cos, we go with case 2.
- If v of x is a polynomial, we go with case number 3.
- D square plus two of y equals x square times a power three x.
- Complementary function the roots are plus or minus root two i because alpha zero beta is root two the complementary function becomes C1 cos root two x plus c two sine root two x.
- You need to go by this logic.
- You have x square into e power 3x.
- If x square is not there, case 1.
- If e power 3x is not there, case 3.
- So now you should go by case number four because both are available.
- times e power 3x.
- This is written as a power 3x times 1 divided by.
- Replace d by d plus a.
- And you can see a values 3.

- and rate it be as it is and observe carefully because it's x square you go by case number three so when you simplify you get case plus 6 d is 11x square.
- It's a polynomial case.
- We want 1 plus x form.
- Then it is written as a power 3x by 11.
- Then apply binomial series we get 1 minus x plus x square minus x power 3.
- Because the highest power of x is 2, we need to go up to d square.
- This is considered.
- This is considered.
- A square is not needed.
- B square is needed.
- 2AB is not needed.
- And in this case, nothing is needed because it is more than 2.
- Then take x square inside.
- Then start up.
- Like differential operator, we get this.
- As usual, the complementary function is going to be c1 plus c2x times e power x because the roots are repeated one and one.
- And this is a particular integrity.
- This denominator equals d minus the whole square so problem becomes simple.
- So it means we need to apply inverse differential operator integrate sine x to times.

Example or application from the lecture

- Let us find out the solution to this differential equation.
- Let us move on to another problem where x is e power x times sine x.

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EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

Core explanation · 15:00–19:59

Answer comes out to be minus a power x times sine x.

Key details

- When x is a polynomial, when x is epo x, epo a x into v, where v is a function of x.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

9 Particular integral of standard functions - Type 4

Study focus: Particular integral of standard functions - Type 4. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000126 · EV0000127

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we are learning a method or a rule to find the particular integral of function of the type a par ax into v where v is any function of x, that is v.

Key details

- We are learning a method to find the particular integral of product of two functions in which one function is exponential function e bar ax.
- Then sine a x plus b or cos e x plus b and algebraic function of the type x to the par n.
- 1 over f of d of e par e x into b is e par x into 1 over f of d plus a of b.
- See e par ax can be taken outside the operator by replacing b by d plus a in f of d.
- What is A here?
- A is the coefficient of x in the exponential function.
- This is the rule used to find the particular integral of product of two functions in which one function is x.
- But in this session, we are solving problems only on type 4, wherein the function is e par ax into v.
- Find the general solution of d2y by dx2 minus 2 dy by dx plus 4y is equal to e to the pi.
- The auxiliary equation is m square minus 2m plus 4 is equal to 0.
- Find out the roots of this quadratic equation.
- By using the quadratic formula, we get the roots as 1 plus R minus.
- Minus root 3i alpha plus r minus i beta alpha is 1 beta is root 3.
- See look at this.
- This is e par x into b where v is equal to cos x.
- E power x can be taken outside the operator by replacement by d plus a, a is one.
- d plus 1 in this e f of d.
- Just replace all d'present in f of d by d plus 1.
- This simplifies to e power x into cos x by d square plus 2d plus 1 minus 2d minus 2 plus 4.
- into cos x by d square plus 3.
- Which rule are we using?
- The second one, cos a x by f of d.
- Cos x by minus 1 square that is minus 1 plus 3, which is e power x into cos x by 2.
- This should be by p.
- Therefore, the general solution is y is equal to yc plus y p.
- This is y p.
- Particular integral when you have product of two functions in which one function is exponential function.

- Product of 3.

Example or application from the lecture

- The given function is of the type e par a x into b.
- Let us consider a problem.
- x into cos x because the given equation is a non-homogeneous LD with constant coefficient, I need to find the complementary function and particular integral.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 05:00–09:59

It is given by x into e power x cos x by d square minus 2d plus 1.

Key details

- into x cos x by d plus 1 the whole square minus 2 into d plus 1 plus 1 which simplifies to this is a plus b the whole square d square plus 2 d plus 1 minus 2 d minus 2 plus 1.
- Plus 2 gets cancelled with minus 2 I get a very simple function of D in the denominator x cos x by d square.
- I know that 1 over d square of any function of x which I call it as capital X is.
- Two times integration of capital X with respect to X.
- I need to integrate this X into cos X with respect to X two times without writing the constant of integration, that's very important.
- without writing the constant of integration.
- I use integration by parts what is integration by parts integral of u into b is u integral b minus integral b into differential of u which you had studied in your 12th standard so it is e to the power x into 1 over d of let me first simplify this is it is x into integration of sine x is minus cos x minus integral of.
- Integration of sin x is minus cos x into differentiation of x with respect to x is 1 dx.
- It simplifies to e to the power x into 1 over d of minus x cos x minus into minus is plus and integration of cos.
- cos x is just sine x.
- We are not writing the constant of integration here.
- How do I integrate this minus x cos x?
- Again by using the same integration by parts minus x into integration of cos x is sine x minus integration of cos x is sine x.
- That sine x when you integrate you get minus cos x.
- Integration of this just sine x is minus cos x.
- This is the particular integral and the general solution is Yc plus YPC.
- Please look here to integrate this x is sine x.
- product of two functions.
- I have used integration by parts.
- You have used integration by parts two times.
- When to integrate x into sin x here and again to integrate this minus of x into cos x.

- This is must y is equal to yc plus y p.
- If the question is to find the general solution of the differential equation.
- Find the general solution of d2 y by dx to minus three dy by dx plus two into y is equal to x into e power 3x plus sine 2x here right hand side I have sum of two functions one is sine a x plus b form for the thirst first term I need to use this rule now for.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

Core explanation · 10:00–14:59

Now let us first write the equation in our notation form, which is d square minus 3d plus 2 of y is equal to x into e power 3x plus sin 2x.

Key details

- Now to find the complementary function, I need to first write the auxiliary equation.
- This is m square minus 3m plus 2.
- equal to 0, find the roots.
- The roots are 1 comma 2.
- So the complementary function is c 1 e to the power 1x plus c2 e to the power 2x y because the roots are real and distinct.
- It is x e to the power 3x plus sin 2x divided by d square minus 3d plus 2.
- First split this into two terms.
- X e to the power 3x by f of d plus sine 2x by f of d where f of d is d square minus 3d plus 2.
- I'll call the first term as particular integral 1 and the second term as particular integral 2, where particular integral 1 is the first term and particular integral 2 is the second term sine 2x by t square minus 3t plus 2.
- Let me first simplify.
- particular integral one and particular integral two and then substitute back in this now to find particular integral one.
- What is it?
- Product of two functions in which one function is e par three x.
- E par three x can be taken outside the operator by replacing b by b plus three in this f of b, so which is [ASR low confidence; check audio].
- x divided by b plus 3 the whole square minus 3 into d plus 3 plus 2.
- X divided by D square plus 3d plus 2.
- Now the next question is how do I simplify this?
- Has many steps.
- What is the first step?
- I need to take the lowest d degree term to get the expression in the form 1 plus 5d or 1 minus 5 d.
- of D where phi F D is D square plus three D by two.
- Take this to the numerator and expand using binomial expansion.
- 1 plus d square plus 3d by 2 whole raise to minus 1 of x.
- Though the tur though the expansion has infinite.

- number of terms because the function on which it has to be operated is x to the power 1.
- The first order derivative of x to the power 1, that is d of x to the power 1 itself is a constant 1.
- When you differentiate it 2, 3, n times the derivatives would be 0.
- So I want only the first derivative d to the power 1.
- I have neglected.
- All higher degree terms.
- In fact, it is 1 minus b square plus 3d by 2 to the power 1.
- Plus d square plus 3d by 2 whole raise to 2.
- I have not considered those terms.
- In the second term also, I have not taken d square by 2.
- I have considered only 3d by 2.
- I need to find particular integral 2 also.
- It's sine 2x divided by d square minus 3d plus 2.
- Which rule I am using over here?
- Rule 2, type 2 type.
- Replace d square by minus into 2 square that is minus 4.
- Retain D term as it is.
- And 2x by 3d plus 2.
- I have a D term.
- I can apply the rule only when I have D square term.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

Core explanation · 15:00–19:59

Convert this d term into d square term by multiply and dividing by its conjugate.

Key details

- What is the conjugate?
- 3d minus 2.
- 2x, which simplifies to 3d minus 2 sin 2x divided by a minus b into a plus b, a square minus b square, 9 d square minus 4.
- Two different ways.
- See I can first apply the operator 1 over 9 d square minus 4 unsine 2x and.
- Apply the operator 3d minus 2 on the resultant function of x.
- First itself you apply the operator 3d minus 2 on sine to x and then you apply the operator 1 over 9d square minus 4 on the resultant function.
- Taken this first 1 over 9 d square minus 4 of sine to x.
- How do I simplify?
- Replace d square by minus into a square.
- What is that minus into a square?
- Minus into 4.
- It is 1 by 4 t into 3d minus 2 of sine 2x.

- Multiply 3d of sine 2x minus 2 of sine 2x.
- What is d of sine 2x?
- It is 2 cos 2x.
- It is PI1 plus PI2, and because the question is to find the general solution, y is equal to complementary function plus particular integral.

Example or application from the lecture

- This is how we find the particular integral when the given function is x into b, where b is a function of x.

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EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

10 Particular Integral when X is any other form

Study focus: Particular Integral when X is any other form. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000142 · EV0000143

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session we learn how to find the particular integral and the right hand side capital X doesn't belong to any of the four standard forms what we have discussed earlier.

Key details

- It means when x is not e power x, when x is not sine of a x plus b or cos of a x plus b, when x is not a polar case, that is x per m, or when x is not e power a x into v case.
- So in this case, we first resolve f of b into linear factors in this form.
- Once we do this, Pi becomes 1 over d minus m1 times d minus m2 and so on, d minus mn.
- Then we need to resolve this into partial fraction.
- This is nothing but e power ax integral e power.
- Minus ax f of x into dx.
- Once we apply this standard formula for each of these n terms, we get a1 e power m1x integral capital X e.
- P power minus m or dx and so on.
- Solve the differential equation.
- T square plus a square of y equals tan ax.
- This is not e power x or sine of cos or polynomial or e power x in to be.
- There's no change in computing the complementary function.
- The roots are plus or minus a i.
- Therefore, the complementary function becomes C1 cos Ax plus C2 sine ax.
- As we know that this f of b should be expressed in this particular form, that is linear factors.

- So based on this idea, these square plus a square can be written as d plus a i times d minus a i it means a plus b into a minus b so once we use partial fraction so this part reduces to one by two ia off of one by d minus i a minus one by d plus i a so probably a and b becomes.

Example or application from the lecture

- Let us consider the differential equation in this particular form.
- We need to apply a standard formula which is given by one over e minus a of f of x.
- Let's take up a problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000142 · EV0000143 · EV0000144 · +5 source units in pack.json

Core explanation · 05:00–09:59

1 by 2 IA and minus 1 by 2 IA.

Key details

- As per the standard formula, this is going to be a power i ax integral tan a x tan a power minus a x into dx.
- And X as it is.
- Which we know very well.
- What is a power plus or minus Ix?
- This is cos x.
- plus or minus i into sin x.
- Therefore e power minus a x becomes cos a x minus i sine a x.
- This tan ax into cos a x.
- After cancellation of cos and cos we get sin a x.
- Then multiply tan ax and sin a x.
- Write sin square x to be one minus cos square ax divided by cos a x.
- I into cos a x of dx.
- Then that leads to this particular step.
- And there's no change in second.
- E power i x times log of secant a x plus tan a x.
- Then that leads to this particular step.
- [ASR low confidence; check audio].
- Then e power iax then as usual log of that particular part plus one by a minus i times e power minus iax.

Example or application from the lecture

- Let us apply the power two times.
- Let's see how to integrate this.
- And let us use Euler's formula.

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EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

Core explanation · 10:00–14:59

into log of that particular part.

Key details

- After doing this simplification, we see that the final PI is going to be.
- Minus 1 by a square log of sec Ax plus than Ax times cos ax.
- Sol the differential equation d square y by dx square e plus log x as you can see it's a special function as compared to those x for standard forms.
- It means zero appears two times.
- So unfortunately we can solve this by a normal method.
- It means log x should be differentiated with respect to x2 times because this is an inverse differential operator.
- So for the first time, as we know, integration of log x.
- should be written as log x times one into dx.
- When we use by parts we get this.
- Finally after the simplification PI becomes x square log x by 2 minus 3x square by 4 and the general solution is.
- Complementary function plus particular integral.
- Again observe this.
- EE power x.
- Then this is the partial fraction.
- A becomes minus one, we become plus one.
- Therefore P i becomes 1 by D plus 1 minus 1 by D plus 2 of E power X.
- So when we use substitution method, this part becomes e power t then dt because e power s dt become dt.

Example or application from the lecture

- Let us solve another problem.
- Let us solve this particular differential equation A square plus 3D plus 2 of y equals EE power x.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

11 Particular integral when X is any other form_Continued

Study focus: Particular integral when X is any other form_Continued. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000158 · EV0000159

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will take up problems of finding particular integral when the right hand side as capital X doesn't belong to any of the four standard forms we have discussed earlier.

Key details

- It means when x is not a power ax, when x is not sine of a x plus b as well as of x plus b.
- When x is not a polynomial case, when x is not e power x into v, where v is a function of x.
- any one of these four forms.
- D plus one times d plus two of y equals sine of v power x.
- This is not any one of these four forms.
- The particular integral is written as 1 divided by f of b which is nothing but d plus 1 times d plus 2 of capital X.
- So we first resolve this into partial fraction.
- Once we use partial fraction.
- This expression reduces to one divided by d plus one minus one divided by d plus two.
- It means a value becomes one, b becomes minus one.
- Therefore, the particular integral is written as 1 divided by d plus 1 minus 1 divided by d plus 2 of this capital X.
- Minus 1 by d plus 2 of sin of v power x.
- We can take this as p2.
- Finally, after computing p1 and p2, we go for p1 minus p2.
- We know a standard formula that is 1 over p minus 1 of f of x equals e power ax integral e power minus ax into f of x to d x.
- So when you apply this formula for this particular case we get a power minus x because area is minus 1.
- Therefore a power minus a x becomes e power minus x.
- A power minus ax becomes a power x, then sine of e power x times dx.
- So we need to integrate this by substitution method.
- Then this expression becomes integral psi t into dt.
- Therefore, the value of P1 is going to be minus a power minus x of cos of e power x.
- One over D plus two of sine of P power X.
- Therefore, as per the standard formula, we get this particular step.
- So when we use substitution method that is a power x equal to t, so that reduces to a power minus 2x integral.

Example or application from the lecture

- It means let us learn how to find out particular integral when the right hand side x is not.
- Let us compute 1 divided by p plus 1 of sine of p power x.
- Let us take e power x equal to t so that dt becomes e power dx.

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EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 05:00–09:59

t s t dt.

Key details

- Therefore, the final PI is going to be P1 minus P2.
- We get this as the answer.
- Provided we have negative sign outside.
- So therefore the particular integral for this problem is minus e power minus 2x into sine of e power x.
- Find the particular integral for the differential equation d square plus phi d plus six of y equals e power 2x times sin of e power x.
- Similar to last problem.
- Finally, Pi can be written as 1 by D plus 2 minus 1 by D plus 3 of E power 2x times sine of E power x.
- In a similar manner, for this case we get e power x times e power 4x times e power.
- E power x into dx.
- After using the same substitution, it reduces to integration of t power four sin t.
- Even this can be solved by integration by partnoli rule.
- The integration of first one is going to be this.
- When we use x t to be e power x, we get this particular step.
- To find out PI when the right hand side X doesn't belong to any of the four standard forms we have discussed earlier.

Example or application from the lecture

- So that leads to minus t cos t plus integral cos t dt because very well how to apply integration by parts let's use the technique so finally we get e power minus 2x of minus t cos t and integral of cos t becomes sine t so finally this is same as e power minus 2x times this.
- In a similar manner, let us solve one more one more problem to get a clarity.
- Let us find the factors of this d plus 3 times d plus 2 of y equals right hand side.

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EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

12 Legendre differential equation

Study focus: Legendre differential equation. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000166 · EV0000167

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We are in fact discussing two different forms of linear differential equation with variable coefficient.

Key details

- The first one being Legendary's linear differential equation.
- And the second one is Cauchy's linear differential equation.
- Cauchy's linear differential equation and methods to solve Legendary's and Cauchy's linear differential equation.
- We learned methods to find the solution of both homogeneous and non-homogeneous linear differential equation with.
- Constant coefficient by suitable substitution.
- See a differential equation of the form.
- Some constant k naught into a x plus b whole power n into n th order derivative of y with respect to x plus k1 into a x plus b whole power n minus 1 into n minus 1 third derivative of y with respect to x.
- plus kn into y where k i is k naught k1 k2 k three k n are constants and capital x is a function of x is called legendary linear differential equation of order n provided k not equal to zero.
- The general form of Legendary's differential equation is of this type.
- K naught into a is one and b is equal to zero in legendary's equation.
- equal to capital X where K0 K1 K2 K3 Kn are constants and capital X is a function of X is Cauchy's linear differential equation of order n.
- Just look at the two differential equations.
- Why I'm calling them as differential equations with variable coefficient because the coefficients of the derivatives are functions of the independent variable x.
- This is an nth order legendary linear differential equation.
- into y is equal to capital X where capital X, the right hand side function, is a function of the independent variable x.
- K0 K1 K2 K3 Kn are all constants.
- E is a legendary linear differential equation of order n.
- Provided this k not equal to 0.
- See ax plus b whole par n into n th order derivative of y with respect to x.
- The degree of the coefficient should be equal to the order of the derivative in all the terms.
- Look at the second term, ax plus b whole power n minus 1.
- The degree of the coefficient is n minus 1.
- And the order of the derivative is n minus 1.
- Look at the last term, it is k kn into y, that is a x plus b to the power 0 into dy by dx to the power 0, which is y.

Example or application from the lecture

- So now let us first learn this legendary linear differential equation in detail.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000166 · EV0000167 · EV0000168 · +1 source units in pack.json

Core explanation · 05:00–09:59

of order n.

Key details

- See the differential equation with variable coefficients is also solved by reducing it to differential equation with constant coefficient.
- Use this differential equation into differential equation with constant coefficient by a suitable substitution.
- We take ax plus b is equal to e power z r, z is equal to log a.
- I'm introducing another variable z which is taken as a function of the independent variable x only.
- Z is equal to log ax plus b.
- This is a substitution I do.
- To convert this differential equation with variable coefficients into differential equation with constant coefficients.
- Let me differentiate it with respect to x.
- A simple differentiation.
- We know what is differentiation of log?
- It is 1 by.
- It is a by ax plus b.
- We should remember this dz by dx is equal to a by a x plus b.
- How did we get this?
- I took the substitution z is equal to log ax plus b.
- Why did I take the substitution z is equal to log ax plus b?
- Because I want to reduce the differential equation with variable coefficients into differential equation.
- equation with constant coefficient.
- It is derivative of y with respect to x.
- Because we have introduced another variable z, I can write d by dz as d by dz into dz by dx.
- How did I write this?
- Using chain rule.
- See because z is a function of x, d by dz can be written as d by dz into dz by dx using the concept over here I get dy by dx as dy by dz into dz by dx.
- Transposing this a x plus b to the left hand side, I get a x plus b into dy by dz is a into dy by dz which is equal to a into d of y where I have called capital D as D by Dz.
- Please note down here.
- See so far I was using the notation capital D for D by DX.
- I can use a different notation because we are familiar with this notation.
- Let me use the notation same notation for D by Dz, but please note here the independent variable over here is Z, it is not X, D by Dz.
- A into d of y.

- Why this z?
- Because now I am changing the independent variable from x to z in the differential equation.
- See that this with the variable coefficient, dy by dx with the variable coefficient.
- is transformed into the constant coefficient by the substitution z is equal to log ax plus b.
- d2 y by dx2.
- What is d2y by dx2?
- It is the derivative of dy by dx with respect to x, which is d by dx of dy by dx.
- Dy by dx.
- It is dy by dz.
- into a by a x plus b.
- Using the above equation here I get a into x plus b into dy by dz.
- How do I simplify this?
- This is a product of two functions.
- What is product rule in depth in differentiation?
- U into v.
- That is u into differentiate differentiation of v plus v into differentiation of u.
- So second function into derivative of first plus first into derivative of second.
- Ah minus a square.
- I differentiated a by a x plus b with respect to a so what do I get?

Example or application from the lecture

- So what I do it I consider this as first function, this is the second function.

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EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

Core explanation · 10:00–14:59

dx of dy by dz.

Key details

- Again I use the same chain rule.
- D by dz can be written as D by dz into dz by DX.
- Minus a square by x plus b the whole square into dy by d dz plus a by a x plus b into d by dz of d y by dz into dz by dx.
- What is dz by dx?
- Which is a by a x plus b.
- I am writing this term first.
- What is it?
- D by dz of dy by dz, meaning it is the differentiation of this dy by dz with respect to z, which is d2 y by dz2 minus dy by dz.
- As a square by a x plus b the whole square into this is capital D is the notation used for D by Dz.
- Minus D of Y.

- I take a x plus b the whole square into d2y by dx2 is equal to a square into d into d minus 1 of y.
- With variable coefficient.
- The derivative with variable coefficient is converted into derivative with constant coefficient, but x is changed to z.
- Please remember it is d by dz, but not d by dx.
- We can prove that a ex plus b whole cube into d3 y by dx3 is equal to a cube into d into d minus 1 into d minus 2 of y.
- Three linear factors.
- D into d minus 1 into d minus 2 of y.
- The order remains the same.
- It is third order.
- Third order here.
- D3y by dx3 is d into d minus 1 into d minus 2 of y, where d is equal to d by dz.
- A x plus b dy by dx is a dy, a x plus b the whole square d 2 y by dx2 is a square into d into d minus 1 of y, a x plus b the whole cube d three, y by d x3 is a cube into d into d minus 1 into d minus 2 of y.
- Using these results in the legendary.
- General differential equation, we get linear differential equation with constant coefficients, which can be solved by the methods discussed so far.
- How are we getting differential equation with constant coefficient?
- Please look here.
- This is KK.
- A k naught into x plus b whole power n into n th order derivative of y with respect s become some constant into d into d minus 1 into d minus 2 so far of y.
- With constant coefficients, which can be solved by the methods learnt before.
- Second order legendary scenear differential equation.
- into a x plus b dy by dx plus k2 of y is equal to capital X.
- The order of the differential equation is 2.
- Ax plus B whole square into the second order derivative of y with respect to x.
- Ax plus b to the power 1 into first order derivative of y with respect to x.

Example or application from the lecture

- Let us now consider legendary's differential equation of order 2.
- Consider so it is a x plus b the whole square, d2 y by dx2 plus k1.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 15:00–19:59

Into dy by dx to the power 0, which is y.

Key details

- And the coefficients are pure functions of x.

- Again, by reducing this differential equation into differential equation with constant.
- coefficients by doing the substitution z is equal to log ax plus b.
- When you put z is equal to log ax plus bc, this a x plus b the whole square d2 y by dx2 becomes a square into d into d minus one of.
- Y plus K1 into A into D of Y plus K2 of Y.
- is equal to is equal to r here you should note that the right hand side was a function of x in the legendary general form.
- Now when you are reducing the differential equation into differential equation with constant coefficient because you are changing the independent variable.
- From X to Z, RHS also should be a function of Z only.
- Differential equation in which Z is the independent variable and Y is the dependent variable.
- Cauchy's linear differential equation.
- Cauchy's linear differential equation of order n.
- See in this differential equation also please note the degree of the coefficient is n and the order of the derivative is n.
- The degree of the coefficient is n minus 1, order of the delimitative is n minus 1.
- But how did we get this Cauchy's linear differential equation?
- It is a special case of legendaries linear differential equation in which a is equal to 1 and b is equal to 0.
- The legendary differential equation was a x plus b whole power n into n th order derivative of y with respect to x.
- Just put a is equal to 1, b is equal to 0.
- We get Cauchy's linear differential equation.
- The same concept is used.
- There in Letch and Grace equals Z is equal to log Ax plus B.
- Here Z is equal to log x.
- When you differentiate, you get bz by dx as 1 by x.
- Where if you do the derivation the same way, we get x into dy by dx as dy by dz, that is d of y where capital D is equal to d by dz.
- I am not doing the derivation for this because it is it is the same way as we did for legendary's differential equation.
- Y where d is equal to d by dz. x cube into d3 y by dx3 would be d into d minus 1 into d minus 2 of y.
- Coefficient, the variable coefficient is converted into a constant coefficient term.
- Substituting these equations in the.

Example or application from the lecture

- Given Cauchy's differential equation, that is the general form of Cauchy's differential equation, the equits reduced to differential equation with constant coefficients, and the equation can be solved by the methods learnt in the previous sessions.

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EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

Core explanation · 20:00–24:59

Cautious differential equation of order two.

Key details

- How does it look like?
- See this is the order of the differential equation is 2.
- and it is with variable coefficient and it is of the form x square into d2 y by dx2 x to the power n 1 into dy by d x to the power 1 x to the power 0 into dy by dx to the power 0.
- Cautious linear differential equation of order 2.
- The same way we do the substitution, z is equal to log x.
- If you do the substitution, z is equal to log x, x into dy by dx becomes d of y where.
- D is equal to d by dz, x square into d2 y by dx2 becomes d into d minus 1 of y.
- If you substitute in the differential equation, the differential equation gets converted into differential equation with constant coefficient.
- See, please note here.
- This function of x also should be transformed to function of z using this substitution.
- Z is equal to log x.
- This is a simple second order linear differential equation with constant coefficients, which can be solved by the methods which we learnt in the previous session.

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EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

13 Legendre differential equations_Continued

Study focus: Legendre differential equations_Continued. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000186 · EV0000187

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will be solving problems on Legendre's linear differential equation and Cauchy's linear differential equation.

Key details

- In the previous session, we discussed about two different forms of linear differential equation with variable coefficients.
- One is legendary's linear differential equation, and the other one is.
- Cauchy's linear differential equation.
- We also learn method to solve Cauchy's and Legendary's linear differential equation.
- A quick recap of the concept studied in the previous session.
- What is this legendary linear differential equation?

- Our differential equation of the form k naught into a x plus b whole par n into nth order derivative of y with respect to x plus.
- K1 into a x plus b whole power n minus 1 into n minus 1 for the derivative of y with respect to x plus so on.
- Plus kn of y is equal to capital X where capital X is a function of the independent variable X, K0, K1, K2, K3, Kn are constants is legendary's linear differential equation of order n.
- See here, the degree of the coefficient is equal to the order of the derivative in all the terms.
- constant coefficients by the substitution z is equal to log a x plus b or e to the power z is equal to a x plus b or e to the power z minus b by a is equal to x.
- In the previous session, we derived that a x plus b into dy by dx is equal to a into d of y by capital.
- D stands for d by d dz.
- It is the differential operator with respect to z because.
- By using this substitution, we are changing the independent variable from X to Z.
- We also derived a x plus b the whole square into d2 y by dx2 is a square into d into d minus 1 of y, and a x plus b the whole cube into d3 y by dx3 is a cube into d into d minus 1 into d minus 2 of y.
- The differential equets transformed into LDE with constant coefficients, and the differential equation is solved by the methods which we learnt in the previous.
- We learned that is Cauchy's linear differential equation.
- We learned that Cauchy's linear differential equation is a special case of.
- Legendary C wherein a is equal to 1 and B is equal to 0.
- derivative of y with respect to x plus kn into y is equal to capital X where K0 K1 K2 K3 Kn are constant and K0 not equal to 0 and the right hand side function is a pure function of X Cauchy's LDE of all.
- The same method is used.
- With variable coefficient is reduced to differential equation with constant coefficients by using the substitution z is equal to log x or x is equal to e power z and x into dy by dx is equal to d of.
- X square where capital D stands for d by dz.
- X square into d2 y by d2 y by dx2 is d into d minus 1 of y.
- X cube d3 y by dx3 is d into d minus 1 into d minus 2 of y.
- Using these equations back in the differential equation, the equation gets transformed into LDE with constant coefficients, and the equation can be solved by the methods which we learned in the previous sessions.

Example or application from the lecture

- Now because in this session we are solving problems on Legendary's LDE and Cauchy's LDE, let us have.
- Using these equations back in the given legendary linear differential equation.
- Now let us apply the method which is learned to solve problems on and legend raised linear differential equation.

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EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

Core explanation · 05:00–09:59

It is LDE with variable coefficients.

Key details

- And we learned only two different forms of LDE with variable coefficients.
- This is Cauchy's linear differential equation of order 3.
- equation by doing the substitution z is equal to log x r x is equal to e power z.
- We also derived x into dy by dx is equal to df y x square capital D stands for d by dz, x square into d2 y by dx2.
- D into d minus 1 of y x cube d3 y by dx is d into d minus 1 into d minus 2 of y.
- Here RHS also has to be converted into a function of z because we are changing the independent variable from x to z.
- X is e power z.
- What is 1 by x?
- It is e power minus z.
- A differential equation in which y is the dependent variable and z is the independent variable.
- Multiply these two terms 2 d square minus 2 d plus 2 of y is equal to 10 into e power z plus e power minus z.
- Simplifying it further, I get dq minus d square plus two of y is equal to 10 into e power z plus e power minus z.
- What is this differential equation?
- It is LD of order 3 with constant coefficient.
- Now, because RHS of the differential equation is not equal to 0, it is a non-homogeneous LD with constant coefficients and of order 3.
- How do I solve this differential equation?
- I know that the non-homogeneous LDE with constant coefficients has two components.
- One is the complementary function, and the second component is the particular integral.
- function and particular integral of this differential equation.
- Now next to find complementary function.
- I need to write the associated auxiliary equation.
- Is it cubic equation three roots now how do I find the roots see just by infect inspection I can find that m is equal to minus one is one root m is equal to minus one because it is minus one.
- And this is minus 1 minus 2 plus 2 equal to 0.
- Once you get one root, the other two roots are found by factorizing.
- The roots are m is equal to minus 1, comma, 1 plus r minus i.
- One is a real root and the other two roots are complex.
- One real root and two complex roots.
- The complementary function y c is equal to c 1 e to the power minus 1 into z.
- Ah, here it is not x because the independent variable in the differential equation is z.

- C1 E power minus Z plus.
- For this now I'm writing P par Z into C2 cos 1 Z plus C3 psi 1 Z, but.
- The complementary function has to be written in terms of x itself.
- 1 by x.
- What is e power plus z?
- It is x.
- Sign of log x.
- See here you should note that the final answer should be in terms of x itself.
- Next is to find the particular integral.
- It is 10 into e power z plus e power minus z divided by d q minus d square plus 2.

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EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

Core explanation · 10:00–14:59

It is 10 into e power z by d cube minus d square plus 2 plus 10 into e power minus z by d cube minus d square plus 2.

Key details

- See, I know that d is equal to minus 1 is a root of dq minus d square plus 2.
- I need to differentiate this, put 1x over here.
- We know that d is equal to 1 is not a root, so the denominator is definitely not equal to 0.
- 1 minus 1 plus 2, which is source 2.
- Now for the second term because the denominator is 0, what I need to do.
- So the first term is 10 e power z by 1 cube minus 1 square plus 2.
- Here I am writing z in 2.
- 10 e power minus z by dq.
- This dq minus d square plus 2 has to be differentiated with respect to d.
- It is 3d square minus 2d.
- See it is z into.
- Please note that it is not x into.
- Because the independent variable here is z, but not x.
- I'll do the substitution d is equal to minus 1.
- I get 10 e power z by 2 plus z into 10 e power minus z by 3.
- into d is equal to minus 1 minus 1 square minus 2 into minus 1.
- The denominator is not equal to 0's.
- Complete solution of the differential equation it is C F plus PI.
- It is it should not be in terms of Z.
- Solve 2x plus 3 the whole square, d2y by dx2 minus 2 into 2x plus 3, dy by dx minus 12y is equal to 6x.
- 2x plus 3 the whole square into second order derivative minus 2 into 2x plus 3 to the power 1 into dy by dx minus 12 into y is equal to 6x is the independent variable.
- See log to the base e.
- We also derived a 2x plus 3, that is a x plus b into dy by dx is e into d of y.

- I need the second one, 2.
- Where capital D stands for D by Dz.
- This has to be written where D is equal to D by C D Z.
- It is not D by DX.
- What is this?
- 4 into d into d minus 1 of y minus 2 into 2.
- The into d of y minus 12 of y is equal to 6 into r. x has to be written in terms of z using this equation.
- E power z minus 3 by 2.
- What is this differential equation?
- It is a linear differential equation with constant coefficients.
- To find complementary function, I need to write the associated auxiliary equation, which is m square minus 2m minus 3 is equal to 0.

Example or application from the lecture

- Let us consider the first return.

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EV0000194 · EV0000195 · EV0000196 · +1 source units in pack.json

Core explanation · 15:00–19:59

To find the roots, the roots are 3 comma minus 1.

Key details

- It is in terms of z.
- But I need to reuse this equation 2x plus 3 is equal to e.
- e power z and write it in terms of x r x is equal to e power z minus three by two.
- How did I write this?
- The numerator 3 to the power z minus 9 is split it is split into two terms.
- Now, how do I simplify the first one?
- It's very simple.
- I just have to replace D by 1.
- Because it is e to the power a is a is 1, just put in the place of d you just put 1.
- Constant by f of d.
- 9 into e power 0z.
- divided by 4 into d square minus 2 d minus 3.
- Replace D by 0.
- D by 1 in the second term I have replaced D by 0.
- So this is because 1 is written as e to d by 0 and here because it is 0 into Z, so it is I have to just replace D by 0.
- But you need to write epar z in terms of s, which is 2x plus 3.
- into 2x plus 3 is the particular integral.
- Please note that when you are reporting the final solution, the complementary function and particular integral should be a function of x, which is the independent.

- Find the general solution.
- It is x into fourth order derivative of y with respect to x plus 10 into third order derivative of y with respect to x is equal to 0.
- By just looking at the differential equation, we say that it's a homogeneous differential equation.
- And but this differential equation doesn't resemble.
- I either Cauchy's or Legendary's linear differential equation, but this equation can be brought to Cauchy's linear differential equation.
- By multiplying the entire differential equation with x to the power 3 y.
- See it is x into 4th order derivative of y with respect to x.
- I want because the order of the derivative here is 4, I want x to the power 4 here.
- And it is 10 into x to the power 0 into third order derivative of y with respect to x.
- Because it is third order derivative, the coefficient of the.
- the degree of the coefficient should be 3.
- See multiplying the entire equation with x cube, it becomes x to the power 4 into d4 y by d.
- X4 plus 10 x cube into third order derivative of y with respect to x is equal to 0.
- Again by doing the substitution.
- x is equal to e to the power z, or z is equal to log x.
- X into d y by dx is equal to d of y where d is equal to this has to be mentioned where capital D is equal to d by dz.

Example or application from the lecture

- This is the complementary function of the given less generalized linear differential equation next is to find the particular integral.
- Variable of the given differential equation.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000198 · EV0000199 · EV0000200 · +1 source units in pack.json

Core explanation · 20:00–24:59

X square into D2 Y by Dx2 is D into D minus 1 of Y.

Key details

- I need up to this four therivative.
- X cube into D3 Y by Dx3 is D into D minus 1 into D minus 2 of Y and X to the power 4 into 4th order derivative of y with respect to X is D into D.
- into d minus 1 into d minus 2 into d minus 3 of y.
- I get this term d into d minus 1 into d minus 2 into d minus 3 of y plus 10 into d into d minus 1 into d minus 2 of y is equal to 0.
- No need to multiply the linear factor.
- What are the common factors?
- fact is d into d minus 1 into d minus 2 can be taken as common fact.
- So I get d into d minus 1 into d minus 2 what is remaining the first term d minus 3 plus 10.
- Of y is equal to 0.

- Simplification would give d into d minus 1 into d minus 2 into d plus 7 of y is equal to 0.
- Now I need to this because this is a homogeneous LDE with constant coefficient and.
- I do I need not have to find the particular integral.
- I need to find only the.
- Complementary function and I know that the complementary function represents the complete solution of the homogeneous linear differential equation.
- The auxiliary equation is m into m minus 1 into m minus 2.
- To m plus 7 is equal to 0.
- The roots are 0, comma 1, comma 2, comma minus 7.
- So I have written I have not written y c I have written y because it is a homogeneous differential equation.
- C4 e to the power minus Z, but the final solution has to be in terms of X itself.
- This is how we find the general solution of Cauchy's and Legendary's linear differential equation.
- So, we learn method to find the general solution of Cauchy's and Legendary's linear differential equation.

Example or application from the lecture

- Let us take a common factor from the two terms.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000202 · EV0000203 · EV0000204 · +1 source units in pack.json

14 Variation of Parameters method_Continued

Study focus: Variation of Parameters method_Continued. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000206 · EV0000207

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we are solving problems on method of variation of parameters.

Key details

- In the previous session, we learned a technique.
- Called method of variation of parameters to solve second order linear differential equations.
- With constant or with variable coefficients.
- Second order linear differential equations are very important in science and engineering.
- They have historically been the most thoroughly studied class of differential equations.
- Find the particular integral by the method of variation of parameters.
- See, we learnt in the previous session that this method of variation of parameters is used to find the particular integral of a differential equation.

- Provided we know the complementary function of the differential equation.
- So even to find the particular integral, we need to first find out the complementary function of the differential equation.
- So this method involves three major steps.
- The first step being finding the complementary function.
- So that is the first step.
- to find by 1 and by 2.
- M square plus 4 is equal to 0.
- Plus r minus 2i.
- The two roots are complex in nature.
- Where alpha is 0 and beta is 2.
- This is this equation is of the form c1 by 1 plus c2 by 2 where y 1 is equal to cos 2x and y2 is equal to sine 2x.
- This is the first step to find y 1 and y2.
- The second step being finding the Ron scale of y 1 and by 2.
- Y1 dash y2 dash.
- Here y1 is cos to x and y2 is sine to x.
- That is the first row.
- And the second row is differentiation of cos to x is minus 2 sine 2x and differentiation of sine 2x is 2 cos 2x.
- Now expand the determinant, it is 2 cos square to x plus 2 sin square 2x, which is equal to 2, because cos square 2x plus sin squared 2x.
- This is 2.
- The Ran scale of y 1 by 2 is 2.
- The third step being finding the particular integral.
- y1 is cos to x and y2 is sine to x.
- It is two, the Roll scan of I1 by two.
- See the major disadvantage or drawback of this method is this method requires evaluating the integrals.
- How do you evaluate this?
- See I can write this as minus sine two x by two.
- Tan 2x is written as sine 2x by cos 2x.
- Sin into sine is sin square 2x by cos 2x dx.

Example or application from the lecture

- Now let us apply the method which is learnt in the previous session to solve problems on second order linear differential equations.
- Now let us apply the method to solve second order differential equations.
- You consider a question.

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EV0000206 · EV0000207 · EV0000208 · +1 source units in pack.json

Core explanation · 05:00–09:59

divided by cos 2x dx.

Key details

- Integral 1 by cos is second secant 2x minus cos 2x dx.
- What is integration of secant 2x?
- It is log of secant 2x plus tan 2x divided by 2.
- And what is integration of cos 2x?
- It is just sine 2x.
- Integration of cos 2x is sin 2x by 2.
- To write the particular integral, we need to find another function called v.
- How do we find v?
- What is y1?
- Y1 is cos 2x.
- W is 2 and capital X is stand to X.
- Tan to x by 2.
- This is a simple integral to evaluate because this is cos to x by 2 and tan is written as sin by cos.
- Cas and cos gets cancelled, so it is half integral sine to x.
- Minus cos 2x divided by 2.
- B is equal to minus cos 2x by 2.
- See, I had told you when we were discussing this method of variation of parameters, when you evaluate the integrals.
- To find u and v, we are not writing the constant of integration.
- But to find the particular integral, first thing is we need to know the complementary function of the differential equation.
- The other thing is we need to evaluate the integrals to find the particular integral.
- Find the general solution by the method of variation of parameters.
- D square minus 1 of y is equal to 2 divided by 1 plus e power x.
- How is this question different from the previous one?
- This is again a second order linear differential equation with constant coefficient.
- Because here RHS function is a little complex function, we need to evaluate the integral.
- First step is to find the complementary function, find out by one by two, second step is to find the Ronskian.
- And third step is to find the particular integral.
- And to find particular integral, we need to find out the functions u of x and v of x by evaluating the integrals.
- Now the first step is to find y and byte.
- The auxiliary equation is m square minus 1 is equal to 0, m square is equal to 1, m is equal to plus r minus square root of 1, which is plus r minus 1, 1 root is plus 1, and the other root is minus 1.
- So because the roots are real and distinct, the complementary function is written as C e to the power 1x plus c2 to the power minus x.
- It is not that you have to write in this order only.
- You can also write c1 e to the power minus x plus c e to e to the power plus x.

- Then y1 becomes e to the power minus x and y2 is equal to e to the power x.
- It doesn't matter.
- The final solution of the differential equation will not change if you interchange y1 and by 2.
- E to the power x, e to the power minus x, that is the first.
- The second step is the derivatives.
- E to the power x or differentiation gives e to the power x.
- E to the power minus x or differentiation gives minus e to the power minus x.

Example or application from the lecture

- So now let us apply this method to solve another question.
- So let us use the method to find the general solution of this differential equation by this method of variation of parameters.

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EV0000210 · EV0000211 · EV0000212 · +1 source units in pack.json

Core explanation · 10:00–14:59

W is equal to minus 2 because e to the power x into e to the power minus x is 1 into minus 1.

Key details

- Ron's can we found out.
- What is the third step?
- Third step is to find the particular integral.
- Y2, I have taken it as e to the power minus x.
- Capital X is 2 by 1 plus E power X divided by W, which we found out it is minus 2 dx.
- 1 by e power x into 1 plus e power x dx.
- Now the next question is how do we evaluate this integral?
- What I do is I just resolve this into partial fractions.
- See this can be written as 1 by e power x.
- Minus 1 by e 1 plus e power x.
- See you take L Cm and check E par x into 1 plus e power x is the L Cm.
- It is 1 by e power x minus 1 by 1 plus e power x dx.
- Now the first term is easy to integrate.
- It is e to the power minus dx.
- Integration of e to the power minus x is minus c to the power minus x.
- So the first term I have integrated, the second one I have written as it is minus integral.
- I can take the common factor.
- E power x in the denominator of this.
- If I take e power x common factor, what do I get?
- 1 by e power x, which is e power minus x plus 1.
- And if I take this e power x to the numerator, it becomes e power minus x by 1 plus e power minus x.
- R I can understand this in a different way.
- Multiply both numerator and denominator by e power minus x.

- How do I solve this integral?
- I'll take t is equal to 1 plus e power minus x.
- Minus e power minus x dx.
- Differentiate this t with respect to x.
- I get dt by dx as minus e power minus x.
- T is equal to 1 plus e power minus x.
- integral becomes its equivalent to it is equivalent to integral.
- E power minus x dx can be replaced by minus dt.
- Divided by the denominator 1 plus e power minus x is written as t.
- Because this integral is of the form some dt by t form.
- Differentiation of the denominator.
- We'll give the numerator.
- It is log p.
- Don't write the constant of integration.
- The value of this integral e power minus x by 1 plus e power minus x dx is minus log of 1 plus e power minus x.
- So that I can evaluate the integral by the substitution method.

Example or application from the lecture

- Denominator if you consider 1 into e power minus x is e power minus x plus e power x into e power minus x is 5.
- Let us solve this integral.
- required to find the particular integral of the given differential equation.

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EV0000214 · EV0000215 · EV0000216 · +1 source units in pack.json

Core explanation · 15:00–19:59

V for this again I need to evaluate another integral.

Key details

- Integral y1 into capital X by W dx.
- Minus 2 and 2 gets cancelled e power x by 1 plus e power dx.
- This is a very simple integrality valid.
- This is again of the form dt by t where t is equal to 1 plus e power x.
- It is log t.
- y p is equal to u into y 1 plus v into y2, we get y p is equal to u e c par x.
- Sorry, y 1 is e power x and u is the function which we found out minus e power x plus log of 1 plus e power minus x.
- Plus b is e power minus x into y 2 is minus log of 1 plus e power x.
- And the general solution is equal to yc plus yp because the question was to find the general solution, not just to find the particular integral.
- Now see this method.
- The major drawback of this method is we need to evaluate the integrals.

- To find the particular integral of the differential equation.
- But see, we apply this method to solve differential equation with constant coefficients.
- Just look at the look at the question.
- It is a differential equation with variable coefficients.
- We know this form of differential equation with variable coefficient.
- This is Cauchy's linear differential equation of order 2.
- I know a method to find the general solution of this differential equation, but because here the question is to find the general solution.
- To apply this method, I first have to find out the complementary function.
- How do you transform this differential equation into differential equation with constant coefficients?
- What is the substitution we do?
- Z is equal to log x r x is equal to e power z.
- X into dy by dx is equal to capital D of Y where capital D stands for D by Dz and X square into D2 Y by D x2 is D into D minus 1 of Y.
- and x to the power 4 is written as e to the power 4 z.
- This differential equation to apply this method of variation of my parameters to solve it.
- I need to write the associated auxiliary equation.
- The auxiliary equation is m square minus 4 m plus 3 is equal to 0.
- Simple quadratic equation.
- What are the roots?
- 3 comma 1 because m minus 3 into m minus 1 are the factors.
- 3 and 1 are the roots.
- The roots are real and distinct.
- Please remember z here, 3z plus c2, e to the power z.
- Now this is of the form c one, y1 plus c2, y2, because I am applying the method.

Example or application from the lecture

- Now let us apply the method to solve second order linear differential equation with variable coefficients.
- Consider a question x square y double dash minus 3xy dash plus 3y is equal to 12x to the power 4.
- By the method of variation of parameters, let us apply method of variation of parameters to find the general solution of this differential equation.

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EV0000218 · EV0000219 · EV0000220 · +1 source units in pack.json

Core explanation · 20:00–24:59

To solve this differential equation, let me continue with the same z itself.

Key details

- At the end we will write back x.
- This z we will write back x.

- Y1 and y2.
- Next you find the.
- X is equal to 12 e to the power 4z because I am applying the method to solve this differential equation.
- Capital X is 12a to the power 4z.
- Y1 is e power 3z and y2 is e power z.
- So this is the first step to find y1 y 2.
- Capital X is twelve e power 4z divided by w is minus 2 into e to the power 4z.
- integral e to the power z d is d z.
- It is e to the power z itself.
- Same thing I have written divided by W is minus 2 to the power 4z.
- This gets canceled, so it is minus 6 integration of e to the power 3z dz.
- It is e to the power 3 z divided by 3 into the 6 so it becomes minus 2 into e to the power 3z.
- See here it is 6 e to the power 4z minus 2 e to the power 4z, which is 4 e to the power 4z.
- Y c is equal to c 1 e to the power 3 z to c2 to the power z.
- We found out y c and y p.
- For the transform differential equations, wherein y was a function of z.
- You get Y C is equal to C1, E par three Z is X cube, plus C two e to the C two.
- into x e power z is x and y p is equal to 4x to the power 4.
- So we learned we used this method to solve linear differential equation with constant as well as with variable coefficients.
- of different forms linear differential equations with constant coefficients, with variable coefficients, homogeneous and non-homogeneous linear differential equations in the sessions.
- In the next session, we will apply the methods which we learn, the concept which we learn.
- Two real world problems.

Example or application from the lecture

- is given by determinant e power 3 z, e power z derivatives 3 e to the power 3 z, e power z.
- Please remember I am applying this method to solve this differential equation, which is equivalent of the given differential equation.
- Which is given by u into y 1 plus v into y 2 where u is given by minus integral y 2 into capital X by W DS.

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EV0000222 · EV0000223 · EV0000224 · +1 source units in pack.json

15 Variation of Parameters method

Study focus: Variation of Parameters method. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000226 · EV0000227

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This method is used to find the particular integral of a linear differential equation whose complementary function is already known.

Key details

- So, this method is used to find only the particular integral of the differential equation.
- To find the particular integral of the differential equation, the complementary function must be known.
- Though this method is very general, we will apply this method.
- To solve a second order linear differential equation with constant coefficient.
- P and Q are constants and capital X is a function of X.
- See, I have considered a simple second order non-homogeneous linear differential equation with constant coefficients.
- I know to find the complementary function of this differential.
- And we know that this c1 by 1 plus c2 y2 represents the complementary function of the differential equation only when y1 and y2 are two.
- Two linearly independent solutions of the associated homogeneous linear differential equation.
- When do we say that y 1 and by 2 are two linearly independent?
- Now, because this c1 y1 plus c2 y2, we have taken it as the complementary function, y1 and y2 have to be the linearly independent functions.
- Y1 by 2 and its derivative y 1 dash by 2 dash.
- That is w is equal to y1 into y2 dash minus y2 into y1 dash.
- Y2 is not equal to 0 if y1 and y2 are two linearly independent functions.
- So because here y1 and y2 are linearly independent, so w is not equal to 0.
- What is capital X?
- What are y1 and y2?
- Two linearly independent solutions of the corresponding.
- Homogeneous differential equation.
- I need to evaluate these two.
- To find the particular integral by this method.
- But when I am writing the expression for u and b, I am not writing the constant of integration because the constant of integration gets observed with the constants of the complementary function when I'm writing the general solution of the differential equation.
- So this method basically involves finding out are evaluating the integrals u and b and finding out this Ronski and W.

Example or application from the lecture

- Let us introduce another term called Ronskeian.
- P i is equal to some u of u a function of x, u of x into y1 plus v of x into y2, where u and v are given by u is equal to minus integral y2 into x by w d and u v is equal to the equal to equal to integral y1 into x by w dx.

- That is the right hand side function of the given differential equation.

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EV0000226 · EV0000227 · EV0000228 · +1 source units in pack.json

Core explanation · 05:00–09:59

non-homogeneous linear differential equation with constant coefficient.

Key details

- This method can also be used to find the particular integral of.
- Our differential equation with variable coefficient.
- If we know the general solution of the associated homogeneous equation, in the sense this method can also be used to find the particular integral of LTE with variable coefficient provided.
- This method can be applied to solve a differential equation of any order that is and th order linear differential equation.
- But we are learning in this session m to solve only LTE of order 2.
- When we are writing, when we are finding the using this method to find the solution of the differential equation, we should note down this.
- If it is not equal to one any other constant number, divide by the coefficient and make the coefficient of d2 y by dx to one.
- What are the demerits of this method?
- See what is the main demerit of this method?
- This method is used to find only the particular integral of the differential equation and.
- To find the particular integral also we need to know the complementary function of the differential equation.
- So the complementary function is absolutely required to apply this method.
- And this method involves evaluation of some integrals, which are sometimes very difficult to evaluate or solve.
- So these are the main two demerits of this method.
- Linear differential equation with constant coefficient and with variable coefficient, but we are applying this method to find the solution of only second order LBE.
- Find the particular integral by the method of variation of parameters.
- It is capital D square minus two d minus three of y is equal to x into e to the power minus x.
- D by dz or it is not d by dz.
- We have we are discussing a different method here, variation of parameter method.
- So it is because the independent variable here is x, so d is d by dx.
- We know to solve this differential equation.
- We know to find the complementary function and also the particular integral of this differential equation which we learnt in the previous session.
- How do you find the particular integral of this equation?
- This is type 4.
- It is e to the power ax into v type where v is a function of x.
- So using that method I can find the particular integral.

- But here the question is to find the particular integral using the method of variation of parameters.
- To find the particular integral of this differential equation, by using method of variation of parameter, I need to write the complementary function of the differential equation.
- If the question was to find the particular integral,.
- To find the particular integral by any method, I need not have to find the complementary function because it is by the method of variation of parameters I need to first find out the complementary function of the differential equation.
- So this method involves three steps.
- The first step being writing the complementary function and finding out by one and by two.
- The second step is to find the RAN scale and the third step is to find u and b.
- function is c 1 e to the power 3x plus c2 to the power minus x.
- C e to the power 3x and e to the power minus x are 2.

Example or application from the lecture

- We know the complementary function of the differential equation, and this method we illustrated by taking the example of second order differential equation.
- The coefficient of d2y by dx2 in the given differential equation has to be 1.
- Now let us apply this method to solve.

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EV0000230 · EV0000231 · EV0000232 · +1 source units in pack.json

Core explanation · 10:00–14:59

linearly independent functions.

Key details

- Derivatives of it three into e to the power three x minus e to the power minus x is the second row.
- two x minus three e to the five.
- e to the power 3x into minus e to the power minus x minus half 3 e to the power 3x into e to the power minus x which simplifies to minus 4 into e to the power 2x.
- The particular next the last step is to find the particular integral.
- Minus what is y 2? e to the power minus x.
- By W which we found out it is minus 4 into E to the power 2x.
- Here you have e to the power minus 2s, which simplifies to 1 by 4 e to the power minus 4s into xpx.
- 1 by 4 into C.
- I have taken x as the first function and I'm integrating e to the power minus 4x.
- integral of integration of this is this minus e to the power minus four x by four into differentiation of this x.
- with respect to x is 1 dx.
- This is just one integral evaluation to find u.
- function b to find the particular integral of the differential equation.

- Integration y 1 into x by w dx.
- What is x? x into e to the power minus x.
- W the wrong scale which we found out it is minus 4 into e to the power 2x.
- E to the power 2x gets canceled with e to the equation.
- by 2x it is minus 1 by 4 into integration of x dx.
- It is x square by 2.
- See I told you when you are calculating u and v no need to write the constant of integration.
- And v is equal to x square by 8.
- See this is the particular solution.
- Particular solution should not contain any constant.
- If the question was to find the complete solution, then you should write the complete solution.
- Y is equal to C plus PI.
- We learn this method to find the particular integral of a second order differential equation with constant coefficients, but this method can be applied to solve linear differential equation of any order with constant coefficient or with variable coefficients.
- So we will solve more problems on this method in the next session.

Example or application from the lecture

- The Ronskiana by one by two is given by determinant e to the power three x e to the power minus x that is the first row.
- The particular integral is given by u into y 1 plus v into y2, where u is e minus integral y2 into x by w dx and b is.
- And v is given by integral y 1 into x by wtx.

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EV0000234 · EV0000235 · EV0000236 · +1 source units in pack.json

16 Applications of higher-order linear differential equations - RLC Circuits

Study focus: Applications of higher-order linear differential equations - RLC Circuits. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000238 · EV0000239

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

What is this L Cr circuit?

Key details

- IFP is measured in the current is measured in amperes and the charge and the capacitor is measured in holes.
- R is the resistance and I is the current.
- E R is equal to R into I.
- E C is equal to Q by C.
- The voltage drop across these three components is equal to the EMF supplied by the battery.
- voltage supplied by the battery is equal to voltage log across the three component that is the sum of these three voltage log is equal to electromotte force.
- VL is l into d by dt, VR is i into i, vc is equal to q by c, where i of p is the current which is flowing to the subdued and q is the charge of the capacity.
- And also we know that i is equal to dq by dt.
- Rate of change in the charge is equal to current.
- Putting this back in the differential equation, the differential equation becomes n into d square q by dt square plus r into d q by dt plus q by c is equal to e for t.
- Here's inductance resistance and capacitance icon.
- This EFT, the EMF which is supplied, is a function of P.
- I need to know, we need to know the charge.
- The current at time t is equal to zero.
- If we know the charge and current at time t is equal to zero, we get an expression for q of p and m of p.
- Find an expression for IFP.
- If R and C circuit with R the resistance, of the resistor which is present in the circuit is eight ohms, the inductance is two h capacitance is 0.15, which is connected.
- a source of voltage E of T is equal to 10 volts, assuming zero charge and zero current when T is equal to zero.
- What is the differential equation which represents the R and C circuit?
- Which is L into D square Q by D T square plus R into D Q by DT plus Q by C is equal to E of T.
- L is equal to 2 degrees.

Example or application from the lecture

- It consists of an inductor of inductance CL capacitor of capacitance C, resistor of resistance are in series with the source of EMF which is given by ESP.
- So to derive a differential equity for the LCI C2, we need to first consider the voltage drops across the three components.
- So first one is the voltage drop across the inductor, which I call it as EL, which is given by L into DIB.

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EV0000238 · EV0000239 · EV0000240 · +1 source units in pack.json

Core explanation · 05:00–09:59

R is equal to 8 ohms.

Key details

- The capacitance is equal to 01 0.15.
- And E of P is equal to it is also a constant, 10 ohms.
- B square Q by DP square plus A T to D Q by D P plus Q by 0.1 is equal to 10.
- Simplify this further.
- Nine point one has one by 10.
- Simplifying the equation further.
- Divide the entire equation by 2.
- 10 Q by 2 is 5Q is equal to 10 by 2.
- is this differential equation it's a non-homogeneous second order linear differential equation with constant coefficients we learn methods to find the solution of a non-homogeneous LB constant coefficients so what is it we know that.
- The complete solution of the non-homogeneous L V has two components.
- One is complementary function, and the other one is the particular.
- Solve the quadratic equation.
- The roots are minus 2 plus i minus 2 minus n.
- The two roots are complex in nature.
- Because there will be self-completed.
- It is e to the power alpha t into c1 cos beta t plus c2.
- Sin beta t in this case alpha is minus 2 and beta is 1.
- Minus 2t, C1 cos t plus C2 sin t.
- This doesn't alone represent the complete solution of the differential equation because the even differential equation is a non-homogeneous L D L D E.
- is a component particular integral.
- So just replace D by 0Y because 5 can be written as 5 into E to the part 0 t.
- Replace D by 0.
- which is equal to 1.
- Minus t 2 t into c1 plus t plus e2 sin t.
- This is the complementary function plus particular integral is 1.
- Q of T and E of T is equal to 0, R T is equal to 0.
- That is F0 is 0.
- Q of 0 is equal to 0.
- Use it in equation 1.
- is equal to e to the power minus 2 into 0 plus c 1 cos 0 plus c2 sine 0 plus 1 sine 0 is 0.
- C is 1.
- P found out c1.
- We'll get dq by t as e to the power minus 2 t into.
- Because it was e to the power minus 2t into C1 cos t plus c2 sin t.
- Product of two functions.
- You must product of t to the power minus 2t.
- Differentiate the second term c1 into cos t differentiation is minus sine t plus c2 into differentiation of sine is cos t.
- Plus differentiate the first term.
- P to the power minus 2p differentiation is minus 2 into e to the power minus 2t.
- Ret in the second term as it is.

- C1 cos t plus c2 sin d.
- But what is dq by dt?
- Which is equal to If P.
- DQ by DT which is equal to I of T is equal to in the place of C1, I have written the value of C.
- I of 0 is equal to.

Example or application from the lecture

- And what else is given?
- function and particular integral of this non-homogeneous linear differential equation which ref which is actually which represents the L Cr circuit which is given in the post genetic.
- Particular integral is given by phi by d square plus 4 d plus phi.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000242 · EV0000243 · EV0000244 · +1 source units in pack.json

Core explanation · 10:00–14:59

I of zero is equal to e to the power minus zero is one.

Key details

- Sin zero is zero plus c two into power zero plus minus two into e to the power zero minus one power zero plus c to sine zero.
- Simplify this you will get the value of c two as minus two.
- Substitute for c1 and c2.
- If t is equal to 5 into e to the power minus 2t sine t.
- linear differential equation to solve problems in physics and engineering, in science and engineering.

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EV0000246 · EV0000247 · EV0000248 · +1 source units in pack.json

17 Application problems on Linear differential equations - Simple harmonic oscillation

Study focus: Application problems on Linear differential equations - Simple harmonic oscillation. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000250 · EV0000251

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Linear differential equations have a wide variety of applications in science and engineering.

Key details

- They are used to model different situations, different phenomena in physics and engineering.
- A object with a mass attached to a vertical string which is oscillating to and fro.
- How these two situations can be modeled by second order linear differential equations.
- We know methods to solve second order linear differential equation.
- We solve the differential equation and we will learn more about the two different phenomena.
- And the restoring force of the spring is pushing it upwards.
- When the two forces are equal, the spring comes to an equilibrium position.
- Or if it is pushed upwards, it starts oscillating up and down.
- In the first diagram, there is nothing happening, a spring is just attached to a support.
- In the second diagram, an object with mass m is attached to a spring which is hung from a level surface.
- So because of the gravity, the mass is pulled downwards, but the restoring force is pushing it upwards and it comes from the same.
- To equilibrium position.
- This is force due to gravity.
- This is because of the mass M.
- And K into S.
- K is the force constant.
- We know it is spring constant.
- K into S is the displacement of the spring from the natural position.
- Because of the mass m mg is equal to case.
- It represents the displacement of the mass from the equilibrium position.
- regulatory motion can be represented by a second order linear differential equation.
- One is the force due to gravity, and the second one is the restoring force.
- We know Hook's law.
- What is from Hooke's Law?
- Hook's Law states that there.
- Restoring force of the spring is proportional to the displacement.
- A restoring force is equal to some constant into S plus X.
- It is a total displacement, and K is the proportionality constant, which is called as the force.
- Constant or the spring constant.
- And negative sign here indicates that the restoring force is in the opposite direction of the displacement.
- We know this.
- It is minus k into s plus x.
- The total the fourth I mean the restoring force acting on the system.
- system is minus k into s plus x.
- What is Newton seconds law?
- The force acting on the system is equal to m into a where a stands for the acceleration.
- M into a now.

- Equating the two, the total forces acting on the system is minus force due to gravity, and the second one is the restoring force.

Example or application from the lecture

- Consider a mechanical oscillator and an electrical oscillator.
- And consider an electrical circuit containing an inductor, capacitor, and resistor in series with a source of EMF.
- Let us see how.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000250 · EV0000251 · EV0000252 · +1 source units in pack.json

Core explanation · 05:00–09:59

A is the acceleration.

Key details

- This mg and k s gets cancelled and it becomes.
- M A is equal to minus k into x.
- But we know that the acceleration is a second order derivative of displacement with respect to time t.
- Putting this back in the equation and simplifying the equation becomes d square x by dt square.
- square plus k by m into x is equal to 0.
- It is a homogeneous differential equation, linear differential equation with constant coefficients.
- The auxiliary equation, which is associated with the differential.
- equation find out the roots and write the complementary function.
- It is actually square root of omega square minus omega square which is plus r minus i omega.
- The two roots are complex.
- It is C1 cos omega T.
- plus C2 sin omega T.
- X of T is equal to C1 cos omega T plus C2 sin omega T.
- What is X of T?
- It is the displacement of the mass from its equilibrium position and this equents a simple harmonic oscillation, simple harmonic motion.
- Introducing another variable phi.
- Put this back and simplify, we get x of t is equal to a into cos of omega t plus phi.
- Where a is equal to square root of C1 square by C2 square and tan phi is equal to minus C2 by C1.
- A represent A which is equal to square root of C1 square by C2 square represent the amplitude of the harm.
- motion and it gives the maximum displacement of the mass from its equilibrium position.
- It is a time taken for one complete oscillation.
- And we'll find out the natural frequency of the harmonic oscillator, which is a reciprocal.

- Of the period.
- It is square root of k by m.
- Equation which we derive for this simple harmonic motion.
- A spring is such that it would be stretched to 6 inches by 12 pound weight.
- Find also the amplitude and the period.
- The weight of the object is 12 pound.
- So, because of the weight, it is stretched to 6 inches.
- S is equal to 6 inches, and W which is equal to Mg is equal to 12 pound.
- We know the differential equation.
- It is d square x by dt square plus k by m x is equal to 0.
- Now first we need to find out this k the force constant k and the.

Example or application from the lecture

- The period of this oscillatory motion is given by 2 pi into square root of m by k.
- Consider see a big question.
- If the weight is pulled down 4 inches below the equilibrium point and given an upward velocity of 2 feet per second, determine the motion of the weight assuming no damping, no forces, external forces acting on the system.

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EV0000254 · EV0000255 · EV0000256 · +1 source units in pack.json

Core explanation · 10:00–14:59

Pass of the object M.

Key details

- K and M.
- We have MG is equal to MG is equal to.
- Then this is the weight of the object 12 pounds.
- or 6 inches is equal to 0.5 feet or 1 by 2 feet.
- It is equal to Mg by S.
- which what is m we have derived it is 3 by 8 into this is acceleration due to gravity 32 divided by the displacement of the mass because of the weight which is equal to 6.
- which is equal to how much?
- We found out Kn.
- Omega is equal to 8.
- General solution C1 cos omega t plus C2 sin omega T.
- Did you understand how we got this?
- Because omega square is equal to 64, omega is equal to 8, and the general solution we wrote.
- It is C1 cos omega t plus C2 sin omega T.
- Now the next question is how do we find the constant C1 and C2?
- To find the constant C1 and C2, we need to use the data.
- To find C1 and C2, we will use the initial condition.
- Sine 0 is 0, cos 0 is 1, c1 is equal to 1 by 3.

- What is velocity? dx by dt.
- We get dx by dt is equal to 1 by 3 into cos differentiation is minus sine into 8 t differentiation is 8.
- But the negative sign, why is this negative sign?
- Opposite direction of the displacement.
- If it is downward velocity, we use the sign plus.
- We'll give positive negative sign.
- C2 is equal to minus 1 by 4.
- Say dx by dt is minus 2 and t is equal to 0.
- This term by and you get C2 into A t is equal to now minus 2.
- It is 1 by 3 cos C8 minus 1 by 4 sin A T.

Example or application from the lecture

- So the weight of the mass, weight of the mass we know it is equal to m into g, where g is the acceleration due to gravity because here it is given in feed.
- Let us write the gravitational mean acceleration due to gravity.
- M into g s is known, s is given from the question.

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EV0000258 · EV0000259 · EV0000260 · +1 source units in pack.json

Core explanation · 15:00–19:59

X of T, it is a function of T time.

Key details

- is 1 by 3 and C2 is minus 1 by 4, simplify amplitude is equal to 5 by 12.
- And also what is the next question?
- Period the motion which is 2 pi by omega.
- But what is omega?
- Omega is equal to we have derived omega is equal to 8.
- This is how we find the period of the motion amplitude by writing the equation for the displacement XFP.

Example or application from the lecture

- The amplitude is given by this is the formula square root of C1 square plus C2 square.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000262 · EV0000263 · EV0000264 · +1 source units in pack.json

18 Source slide deck: Solutions of Homogeneous DE by introducing Complementary Functions

Study focus: Source slide deck: Solutions of Homogeneous DE by introducing Complementary Functions. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator.

EV0000320 · EV0000321

SHORT-NOTE RECAP

Study notes · Source slide deck: Solutions of Homogeneous DE by introducing Complementary Functions

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

- Department of Science and Humanities

Slide 2 · ENGINEERING MATHEMATICS I

- SUB-TOPIC: LDE with constant coefficients
- Department of Science and Humanities

Slide 3 · ENGINEERING MATHEMATICS I

- To Solve a Higher Order Homogeneous Linear Differential Equation with constant coefficients
- Case 1. When the Auxiliary Equation has Real and Distinct Roots
- Case 2. When the Auxiliary Equation has Real and Equal Roots

Slide 4 · ENGINEERING MATHEMATICS I

- GENERAL SOLUTION OF SECOND ORDER LINEAR HOMOGENEOUS DIFFERENTIAL EQUATION

Slide 5 · ENGINEERING MATHEMATICS I

- GENERAL SOLUTION OF SECOND ORDER LINEAR HOMOGENEOUS DIFFERENTIAL EQUATION

Slide 6 · ENGINEERING MATHEMATICS I

- GENERAL SOLUTION OF SECOND ORDER LINEAR HOMOGENEOUS DIFFERENTIAL EQUATION

Slide 7 · ENGINEERING MATHEMATICS I

- GENERAL SOLUTION OF SECOND ORDER HOMOGENEOUS LINEAR DIFFERENTIAL EQUATION

Slide 8 · ENGINEERING MATHEMATICS I

- GENERAL SOLUTION OF SECOND ORDER LINEAR DIFFERENTIAL EQUATION

Slide 9 · ENGINEERING MATHEMATICS I

- GENERAL SOLUTION OF SECOND ORDER LINEAR

Slide 10 · ENGINEERING MATHEMATICS I

- GENERAL SOLUTION OF HOMOGENEOUS LINEAR DIFFERENTIAL EQUATION OF ORDER “n”

Slide 11 · ENGINEERING MATHEMATICS I

- GENERAL SOLUTION OF HOMOGENEOUS LINEAR DIFFERENTIAL EQUATION OF ORDER “n”
- then the general solution is

Slide 12 · ENGINEERING MATHEMATICS I

- GENERAL SOLUTION OF HOMOGENEOUS LINEAR
- DIFFERENTIAL EQUATION OF ORDER “n”

Slide 13 · ENGINEERING MATHEMATICS I

- GENERAL SOLUTION OF HOMOGENEOUS LINEAR
- DIFFERENTIAL EQUATION OF ORDER “n”

Slide 14 · ENGINEERING MATHEMATICS I

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 15 · [Image text, OCR]

- Department of Science and Humanities

EV0000320 · EV0000321 · EV0000322 · +42 source units in pack.json

19 Historical question-source pages · unit mapping candidates

Study focus: Historical question-source pages · unit mapping candidates. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000824 · EV0000825

SHORT-NOTE RECAP

Study notes · Historical question-source pages · unit mapping candidates

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Archive page 3 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.5; source text/OCR may be partial. Bessel Differential Equation ( ) 2 2 2 0 x y xy x n y ⊠⊠ ⊠ + + − = is known as Bessel differential equation. ENGINEERING MATHEMATICS I The second order linear ordinary differential equation in the form, (1) −−−− Here n is the order of the Bessel differential equation.

Archive page 14 from Math_Unit2+Unit3+Unit4__1w-kU1CiDL.pdf; unit match candidate score 0.5; source text/OCR may be partial. W - 4 · Non-homogeneous PDE · Linear PDE but order of all P . D aren't equal en : (D2-(D'l 2 + D - D'S z = 0 · Working : · Finding CF , convert F(D , D') into linear factors of the form () D - m , D' - a ,) . (D - myD' -az) .... (D - mpD' - an))z = 0 (F = e* y , (y + m ,x) + ea2y(y + myn) +... + eannyn(y + mnn) a , 92, ... an & m , M2 ... Mn are constant · (D - mD - a) = = 0 p - my - az = 0 Pr + Qq = R After converting du =dy= into linear factors I directly write in -m(dn =dy J& Feder this form y + ma = C logz = <n + log(2 Ch z = 2 . 9 [= f(x) = J(y + mu) z = f(y +mu) . e L · Easy way to factorize non-homogeneous PDE Assume (a, D + b , D' +c)(a2D + b2D +x) en : (2D"-DD' - D + D - D') z coeft of D2 = a , az = ezn +3y coett of D' = b , b2 a, ac = 2 constant = <[2 b, bz = - a : 1 &2 : 2 coeft of D = a , + ac C , 2 = 0 b. : - 1 be : 1 coeft of D' = b , 2 + b 22 a ,x + a24 = 1 c : 0 22 : I b ,c + b24 = - inen, (D-D'S (2D - D' + 1) = (D - D')(D - +z) if same m and a

EV0000824 · EV0000825

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.

Source	Path	Use
Introduction to higher order differential equations, complementary function and particular integral transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/001_Introduction_to_higher_order_differential_equations_complementary_function_and_particular_integral__069274588a013075.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to higher order differential equations, complementary function and particular integral transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/001_Introduction_to_higher_order_differential_equations_complementary_function_and_particular_integral__069274588a013075.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to higher order differential equations, complementary function and particular integral transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/001_Introduction_to_higher_order_differential_equations_complementary_function_and_particular_integral__069274588a013075.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9986; only the JSON segmentation supplies the displayed timestamps.
Introduction to higher order differential equations, complementary function and particular integral transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/001_Introduction_to_higher_order_differential_equations_complementary_function_and_particular_integral__069274588a013075.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Rules for finding the Complementary Function, Problems transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/002_Rules_for_finding_the_Complementary_Function_Problems__4a39dee530243163.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Rules for finding the Complementary Function, Problems transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/002_Rules_for_finding_the_Complementary_Function_Problems__4a39dee530243163.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Rules for finding the Complementary Function, Problems transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/002_Rules_for_finding_the_Complementary_Function_Problems__4a39dee530243163.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9989; only the JSON segmentation supplies the displayed timestamps.
Rules for finding the Complementary Function, Problems transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/002_Rules_for_finding_the_Complementary_Function_Problems__4a39dee530243163.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 1 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/003_Particular_integral_of_standard_functions_-_Type_1__22d81006814d93bf.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 1 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/003_Particular_integral_of_standard_functions_-_Type_1__22d81006814d93bf.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral of standard functions - Type 1 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/003_Particular_integral_of_standard_functions_-_Type_1__22d81006814d93bf.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9990; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 1 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/003_Particular_integral_of_standard_functions_-_Type_1__22d81006814d93bf.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral_Type 1 and Type 2 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_Type_1_and_Type_2__54bec5745a32a447.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral_Type 1 and Type 2 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_Type_1_and_Type_2__54bec5745a32a447.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral_Type 1 and Type 2 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_Type_1_and_Type_2__54bec5745a32a447.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9979; only the JSON segmentation supplies the displayed timestamps.
Particular integral_Type 1 and Type 2 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_Type_1_and_Type_2__54bec5745a32a447.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral of standard functions - Type 2 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_of_standard_functions_-_Type_2__d25d020162de0016.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 2 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_of_standard_functions_-_Type_2__d25d020162de0016.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 2 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_of_standard_functions_-_Type_2__d25d020162de0016.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 2 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_of_standard_functions_-_Type_2__d25d020162de0016.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 3 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/005_Particular_integral_of_standard_functions_-_Type_3__1f8ea5ed4d7e088f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral of standard functions - Type 3 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/005_Particular_integral_of_standard_functions_-_Type_3__1f8ea5ed4d7e088f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 3 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/005_Particular_integral_of_standard_functions_-_Type_3__1f8ea5ed4d7e088f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9983; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 3 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/005_Particular_integral_of_standard_functions_-_Type_3__1f8ea5ed4d7e088f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_-Type_3_and_Type_4__fb358573adb09195.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_-Type_3_and_Type_4__fb358573adb09195.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral -Type 3 and Type 4 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_-Type_3_and_Type_4__fb358573adb09195.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_-Type_3_and_Type_4__fb358573adb09195.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 4 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_of_standard_functions_-_Type_4__603379d034480436.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 4 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_of_standard_functions_-_Type_4__603379d034480436.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 4 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_of_standard_functions_-_Type_4__603379d034480436.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9988; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral of standard functions - Type 4 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_of_standard_functions_-_Type_4__603379d034480436.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Particular Integral when X is any other form transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_Integral_when_X_is_any_other_form__653ae8f75a03fb1e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular Integral when X is any other form transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_Integral_when_X_is_any_other_form__653ae8f75a03fb1e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular Integral when X is any other form transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_Integral_when_X_is_any_other_form__653ae8f75a03fb1e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9975; only the JSON segmentation supplies the displayed timestamps.
Particular Integral when X is any other form transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_Integral_when_X_is_any_other_form__653ae8f75a03fb1e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral when X is any other form_Continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_integral_when_X_is_any_other_form_Continued__5c24666421028690.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral when X is any other form_Continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_integral_when_X_is_any_other_form_Continued__5c24666421028690.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral when X is any other form_Continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_integral_when_X_is_any_other_form_Continued__5c24666421028690.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9967; only the JSON segmentation supplies the displayed timestamps.
Particular integral when X is any other form_Continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_integral_when_X_is_any_other_form_Continued__5c24666421028690.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equation__04e8c1e6bb4bea05.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Legendre differential equation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equation__04e8c1e6bb4bea05.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equation__04e8c1e6bb4bea05.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equation__04e8c1e6bb4bea05.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equations_Continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equations_Continued__e286e5075a9d86f4.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equations_Continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equations_Continued__e286e5075a9d86f4.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equations_Continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equations_Continued__e286e5075a9d86f4.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Legendre differential equations_Continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equations_Continued__e286e5075a9d86f4.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method_Continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method_Continued__97963caf81f64a9c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method_Continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method_Continued__97963caf81f64a9c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method_Continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method_Continued__97963caf81f64a9c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method_Continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method_Continued__97963caf81f64a9c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method__f1407a459331b120.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Variation of Parameters method transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method__f1407a459331b120.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method__f1407a459331b120.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method__f1407a459331b120.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Applications of higher-order linear differential equations - RLC Circuits transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/014_Applications_of_higher-order_linear_differential_equations_-_RLC_Circuits__2d0217a8ab77a9c8.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Applications of higher-order linear differential equations - RLC Circuits transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/014_Applications_of_higher-order_linear_differential_equations_-_RLC_Circuits__2d0217a8ab77a9c8.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Applications of higher-order linear differential equations - RLC Circuits transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/014_Applications_of_higher-order_linear_differential_equations_-_RLC_Circuits__2d0217a8ab77a9c8.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9976; only the JSON segmentation supplies the displayed timestamps.
Applications of higher-order linear differential equations - RLC Circuits transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/014_Applications_of_higher-order_linear_differential_equations_-_RLC_Circuits__2d0217a8ab77a9c8.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Application problems on Linear differential equations - Simple harmonic oscillation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/015_Application_problems_on_Linear_differential_equations_-_Simple_harmonic_oscillation__9f24b70546186809.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Application problems on Linear differential equations - Simple harmonic oscillation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/015_Application_problems_on_Linear_differential_equations_-_Simple_harmonic_oscillation__9f24b70546186809.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Application problems on Linear differential equations - Simple harmonic oscillation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/015_Application_problems_on_Linear_differential_equations_-_Simple_harmonic_oscillation__9f24b70546186809.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9983; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Application problems on Linear differential equations - Simple harmonic oscillation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/015_Application_problems_on_Linear_differential_equations_-_Simple_harmonic_oscillation__9f24b70546186809.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Introduction to second and higher order DE with constant coefficients	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Introduction to second and higher order DE with constant coefficients	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Introduction to second and higher order DE with constant coefficients (converted PDF; original hash 136d74b8e9f37faca1033756c92ffe840c372acf9410d1f1aa49c4b0f1c7a135)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Solutions of Homogeneous DE by introducing Complementary Functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Solutions of Homogeneous DE by introducing Complementary Functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Solutions of Homogeneous DE by introducing Complementary Functions (converted PDF; original hash 080c29fbcaa2829cff08b42036ee05fea171c295f98bf51470eeb038eb890d8a)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral of Type 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Particular integral of Type 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral of Type 1 (converted PDF; original hash 84f8f81caf6c9c435e20a3d93ba55b4f658236fe7202cd78cec415f091aaeea7)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Particular integral of Type 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Particular integral of Type 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral of Type 2 (converted PDF; original hash e2030fff59bf8246db5254db42fc95be9243c269abf0afa18169d12af58417a7)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral of Type 3	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Particular integral of Type 3	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral of Type 3 (converted PDF; original hash a4e3b73e36d4d5c9a4e485bf34097ac2c7093d9ad3f92951f157fd54ecda9e30)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Particular integral of Type 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Particular integral of Type 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral of Type 4 (converted PDF; original hash acefecfd876a5d49e837f0aff6ec3f919809c4b92bd183f728c942051c82f0bc)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral when X is any other form	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Particular integral when X is any other form	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral when X is any other form (converted PDF; original hash d00a5ae64546151ed598e60717523153e6f0bf0d300c6850837c25310f4ff4fb)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Legendre s differential equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Legendre s differential equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Legendre s differential equation (converted PDF; original hash 38cc6e95d1aeca7f1625a42af8612f370d1be985622142a7a86be4b2b3c74f9f)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Variation of Parameters method	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Variation of Parameters method	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Variation of Parameters method (converted PDF; original hash 7cacd33b88c550da86eed1eba7748a51cfa85706f05516eb85d299dbfe05cca4)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Applications of DE LCR circuit	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Applications of DE LCR circuit	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Applications of DE LCR circuit (converted PDF; original hash e3adc11eb94d5de98210fd2f254c966dc6692cab0f8977b9f66d27597f90cc6b)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Applications of DE SHM	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Applications of DE SHM	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Applications of DE SHM (converted PDF; original hash 7b58c61a57ed91331a4fd457402244c66823d50388e69eb1aefecfae82bf6bfc)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Extracted historical pages: Series Solution of Bessel__1GK7DgclHD.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Math 1/Extra/Extra Questions/Series Solution of Bessel__1GK7DgclHD.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: Math_Unit2+Unit3+Unit4__1w-kU1CiDL.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Math 1/Extra/Revision Notes/Math_Unit2+Unit3+Unit4__1w-kU1CiDL.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.

Source	Unit counts	Example decisions
SRC00002	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00003	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00004	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00005	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00006	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...
SRC00007	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...
SRC00009	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00010	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00011	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00013	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00014	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0046. Sidecar variants compare...
SRC00015	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0046. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0046. Sidecar variants compare...
SRC00017	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0046. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00018	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00019	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00021	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00022	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0050. Sidecar variants compare...
SRC00023	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0050. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0050. Sidecar variants compare...
SRC00025	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0050. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00026	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...
SRC00027	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00028	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...
SRC00029	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00030	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00031	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00032	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00033	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00034	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0054. Sidecar variants compare...
SRC00035	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00036	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0054. Sidecar variants compare...
SRC00037	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00038	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare...
SRC00039	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare...
SRC00040	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00041	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare...
SRC00042	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00043	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00044	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00045	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00046	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00047	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00048	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00049	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00050	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00051	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00052	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00053	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00054	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00055	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00056	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00057	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00058	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00059	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...
SRC00060	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00061	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...
SRC00062	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00063	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0056. Sidecar variants compare...
SRC00064	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00065	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0056. Sidecar variants compare...
SRC00066	0 covered · 0 excluded · 18 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00068	0 covered · 0 excluded · 18 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00067	0 covered · 0 excluded · 18 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00069	15 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00071	15 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00070	15 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00072	0 covered · 0 excluded · 18 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00074	0 covered · 0 excluded · 18 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00073	0 covered · 0 excluded · 18 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00075	0 covered · 0 excluded · 18 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00077	0 covered · 0 excluded · 18 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00076	0 covered · 0 excluded · 18 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00078	0 covered · 0 excluded · 15 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00080	0 covered · 0 excluded · 15 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00079	0 covered · 0 excluded · 15 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00081	0 covered · 0 excluded · 15 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00083	0 covered · 0 excluded · 15 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00082	0 covered · 0 excluded · 15 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00084	0 covered · 0 excluded · 18 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00086	0 covered · 0 excluded · 18 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00085	0 covered · 0 excluded · 18 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00087	0 covered · 0 excluded · 11 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00089	0 covered · 0 excluded · 11 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00088	0 covered · 0 excluded · 11 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00090	0 covered · 0 excluded · 18 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00092	0 covered · 0 excluded · 18 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00091	0 covered · 0 excluded · 18 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00093	0 covered · 0 excluded · 17 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00095	0 covered · 0 excluded · 17 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00094	0 covered · 0 excluded · 17 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00096	0 covered · 0 excluded · 23 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00098	0 covered · 0 excluded · 23 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00097	0 covered · 0 excluded · 23 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00099	1 covered · 0 excluded · 0 gaps	q3: Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
SRC00100	1 covered · 0 excluded · 0 gaps	q14: Original archive PDF page 14; candidate unit association is text-based and unconfirmed.

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